

Incentive Compatibility and Payoff Robustness in Multiplayer Quantum Games: Theory and Hardware Evidence

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Abstract—We study how entangler family, player count, and implementation noise affect payoff and incentives in an N -player Eisert–Wilkens–Lewenstein allocation game. Five families define an exact simulation landscape, with graph-role asymmetry distinguished from circuit-order and gate-budget effects. For the global GHZ operator, we derive a family of cooperative phase strategies and an exact restricted-menu incentive boundary. The original branch Q_N , with phase π/N , fails the Hawk-deviation test for $N \geq 4$, whereas an alternative branch Q_N^* , which coincides with Q_N for $N \leq 3$, is a strict restricted-menu equilibrium for every $N \geq 2$ at maximal entanglement. Both admit an explicit profitable unrestricted SU(2) deviation. A four-product-state representation permits exact classical payoff evaluation through $N = 128$ without dense state enumeration. IBM Heron r2 experiments evaluate Q_N through seven players, including preregistered Hawk-deviation tests at $N = 3$ and failed out-of-sample noise-model predictions. Two all-player hardware batches of Q_N^* support the preregistered incentive criterion at $N = 4, 6$ in both runs, but fail the overall five-size test; $N = 7$ remains unresolved. Mean-payoff retention is intrinsically forgiving: uniformly random outcomes retain 99.22% of the ideal analytic-baseline gap at $N = 7$, and measured retention at $N = 6, 7$ falls below that comparator. A retrospective sensitivity analysis that charges for partial conflicts gives raw $N = 7$ retention of 93.23%, against 75% for uniformly random outcomes. These results separate scalable welfare evaluation from incentive compatibility and physical implementation; the trading interpretation is illustrative rather than an economic validation.

Index Terms—quantum games, Nash equilibrium, EWL protocol, entanglement topology, NISQ, error mitigation, quantum finance

I. INTRODUCTION

Financial markets are coordination problems with adversarial payoffs. When traders compete over a scarce asset, each chooses between an aggressive position (capturing the full value if unopposed) and a cooperative one that shares value reliably. The Hawk–Dove game formalizes this tension. With the parameters used throughout this work, $V = 4$ and $C = 3$, Hawk strictly dominates Dove in the two-player payoff matrix: it pays more against either opposing

action, so mutual Hawk is the unique classical pure Nash equilibrium. Because $V > C$, this is the dominance regime of the Hawk–Dove family, with Prisoner’s-Dilemma structure rather than the mixed equilibrium of the textbook $V < C$ case. Yet mutual Dove pays each player $V/2 = 2$, whereas mutual Hawk pays only $(V - C)/2 = 0.5$. The resulting gap isolates the mechanism-design problem: individually rational aggression destroys value that cooperation would preserve. Classical remedies are external to the game, including trust between counterparties, a trusted intermediary, or regulation that taxes defection.

Quantum game theory offers a different lever. In the Eisert–Wilkens–Lewenstein (EWL) protocol [1], strategies live in SU(2) and players’ qubits are entangled before any move is made; the correlation structure of the shared state then changes what is self-enforcing. Within its restricted strategy space, the two-player result is a quantum Nash equilibrium that is simultaneously symmetric and Pareto-optimal within the stipulated protocol. Preparation, joint decoding, and strategy-menu enforcement remain mechanism assumptions. Carbon-trading and order-book architectures provide motivating analogies; we do not establish replacement of regulatory or trust mechanisms.

Khan et al. [2] reported multiplayer ion-trap trading-game experiments with three to six players under a Prisoner’s-Dilemma payoff rule. Our extension concerns a different allocation rule and controlled comparisons of entangler families, incentives, and implementation effects. Real markets are networks: the *topology* of who is correlated with whom is a design variable, and each topology induces a different interference structure, different equilibria, and a different response to decoherence. Varsamis et al. [3] studied N -player entanglement operators and scalability on IBM hardware; our contribution is orthogonal: we fix a financially interpreted game and chart how *topology* $\times$ *player count* $\times$ *noise* determine both the magnitude and the equilibrium status of the quantum advantage, then test the resulting noise model against

preregistered predictions on hardware.

The main contributions of this work are as follows:

- 1) **Incentive boundary and scalable evaluation.** We derive the GHZ cooperative phase branches and their exact restricted-menu deviation condition. An alternative branch Q_N^* , identical to the original Q_N for $N \leq 3$, satisfies that condition for every $N \geq 2$ at maximal entanglement, while an explicit unrestricted SU(2) deviation defeats it. A four-product-state representation enables exact classical payoff evaluation through $N = 128$, distinct from physical player-count scaling.
- 2) **Advantage landscape.** The complete $(N, \text{topology})$ circuit-relative payoff-gap map for $N = 2-6$ over five topologies. For GHZ, the fixed cooperative profile $(Q_N, \dots, Q_N)$ pays exactly V/N and passes the restricted $\{D, H, Q_N\}$ deviation check at $N = 2, 3$, not against arbitrary SU(2) gates; ring and fully connected also pass that restricted check at $N = 3$. Other topology cells may or may not attain V/N .
- 3) **Noise-robustness dichotomy.** Under depolarizing noise the restricted-menu equilibrium criterion and circuit-relative payoff-gap criterion order topologies differently; noise also breaks player symmetry, producing position-locked per-player disadvantage that follows the entangler's gate ordering (shown by a wiring-permutation experiment). Matched-gate-count and compilation controls reduce gate-budget confounding within the XX-product family (GHZ/ring/star/fully connected), whose normalized decay is similar (Table III), while absolute GHZ-W ordering can remain because their zero-noise payoff gaps differ. W is the normalized-retention outlier at approximately 0.8–1.0 per CX. In an exploratory W $N = 4$ finite-round pilot, no tested setting produced a beneficial converged fairness repair.
- 4) **Preregistered hardware tests.** On `ibm_fez`, the original-phase GHZ protocol at $N = 3, 4, 5$, with tensored readout mitigation and CZ-fold zero-noise extrapolation, retains 98–99% of the ideal advantage $3/N$, above the 87.5–96.9% that uniformly random outcomes would retain. A preregistered one-parameter depolarizing model fitted at $N = 3$ misses the $N = 4, 5$ endpoints; across three repeats the $N = 4$ prediction passes once, and the $N = 5$ deficit recurs in every repeat. An $N = 3-7$ extension retains 96–99% but falls below the random-outcome comparator at $N = 6, 7$. Two all-player batches of Q_N^* , testing every Hawk and Dove deviation, support the preregistered incentive criterion at $N = 4, 6$ in both batches but fail the overall five-size test. We identify why the mean payoff decays more slowly than the target-state population: it is first-order insensitive to single bit-flips.
- 5) **Reproducible pipeline.** Open simulation and hardware code that checks every circuit against its dense reference and rehearses each batch on a device noise model before submission.

The rest of the paper is organized as follows. Section II reviews related work, and Section III defines the evaluation questions and evidence boundaries. Section IV develops the mathematical framework, followed by computation and validation methods in Section V. Section VI reports simulation and hardware results. Section VII discusses their interpretation, scalability, and limitations. Section VIII states the technical differentiation, followed by data and code availability and the conclusion in Section IX.

II. RELATED WORK

A. Quantum games and the EWL protocol

The EWL quantization [1] embeds player strategies in SU(2) and entangles the player register before any strategy is applied. Its equilibrium properties depend on both the entangler and the permitted strategy set. The original two-player restricted-strategy result admits a symmetric, Pareto-optimal equilibrium at maximal entanglement, $\gamma = \pi/2$; enlarging the strategy space can invalidate that equilibrium [1, 4]. The protocol is the base layer of this work: we keep its structure and vary what it entangles.

B. Multiplayer extensions and entanglement structure

Analytic multiplayer cooperation equilibria also occur in the quantum volunteer's dilemma [5]. Its payoff and classical comparator differ from the allocation game studied here; an extension to multiple players alone is therefore not our novelty claim. Benjamin and Hayden [6] give a multiplayer quantum-game construction, including minority-game and Prisoner's-Dilemma examples. We adopt its circuit framework; the shared-resource allocation payoff in Sec. IV is the separately specified model studied here. Chappell, Iqbal, and Abbott [7] compared GHZ and W entanglement in N -player quantum games in an EPR setting, but in simulation and without a financial payoff structure or a hardware validation; our contribution is to carry that GHZ-versus-W question through a concrete trading game to a noise-model-tested result on a real device.

C. Quantum advantage in trading games

Khan et al. [2] developed two-player trading models and reported multiplayer ion-trap experiments with three to six traders using a Prisoner's-Dilemma payoff rule. We distinguish our allocation game and entangler-family controls from that prior multiplayer hardware demonstration. Within the simple pairwise graph convention, two players admit only one edge: the question of which correlation graph a market designer should choose only appears at $N \geq 3$, and that is the question this paper addresses.

D. N -player hardware scalability and error mitigation

Varsamis et al. [3] study N -player entanglement operators and their scalability on IBM hardware generically, characterizing the operators themselves across devices. We instead fix

a single game (the financial Hawk–Dove) and map its *advantage landscape*: the topology-dependent equilibrium/payoff dichotomy, the noise-induced position-locked unfairness, and a preregistered predictive noise model whose out-of-sample predictions we test on hardware. On the mitigation side we adopt two standard NISQ techniques rather than proposing new ones: tensored confusion-matrix inversion for readout errors [8], [9] and zero-noise extrapolation by local gate folding [10], [11]; their use in this pipeline is specified in Sec. VI-B.

III. EVALUATION QUESTIONS AND EVIDENCE BOUNDARIES

The study addresses four questions with different units of evidence. First, which cooperative phase profiles resist unilateral deviations in the specified game? Second, how do entangler families change payoffs and fairness at a fixed strategy? Third, which apparent robustness differences survive controls for circuit implementation? Fourth, which ideal properties remain supportable on a finite noisy processor? Table I states the operational test and interpretation of each question.

A. Mathematical identities and empirical estimands

The phase boundary and four-product-state representation are algebraic properties of the ideal GHZ protocol. Finite floating-point tests check their implementation but do not establish the all- N statements by themselves. Hardware probabilities instead estimate repeated circuit outcomes. The mean payoff, each player’s payoff, the worst unilateral gain, and the all-zero population are distinct estimands. In particular, high mean payoff cannot substitute for a player-level incentive test.

B. Evidence hierarchy and negative findings

Exploratory runs, retrospective reweighting, and preregistered tests are reported separately, and failed predictions are reported alongside passes. The alternative-phase repeat is judged under the same rule as the first batch, without pooling to reverse a failed acceptance test. No result on a finite device is promoted to a statement about unlimited physical scaling. Compact classical evaluation is reported as an analysis tool, while physical player count remains the number of simultaneously represented qubits in the executed protocol.

IV. PROPOSED N -PLAYER QUANTUM HAWK–DOVE FRAMEWORK

A. Notation and the Classical N -Player Payoff Tensor

Before quantizing the game, we need a payoff rule that assigns a value to every one of the 2^N classical outcomes and preserves each player’s identity.

We write $\mathcal{N} = \{1, \dots, N\}$ for the set of players. Every player $j \in \mathcal{N}$ chooses one of two actions, and we record a joint choice as a bitstring $x = (x_1, \dots, x_N) \in \{0, 1\}^N$, where $x_j = 1$ means player j plays Hawk and $x_j = 0$ means player j

plays Dove. Hawk is the aggressive action: it takes the whole contested value when unopposed, but destroys value when it meets itself. Dove is the cooperative action, which shares value reliably. Two constants set the stakes: the contested resource is worth $V > 0$, and a confrontation between Hawks costs $C > 0$. This is the two-strategy value-and-cost structure of the classical Hawk–Dove game [12], carried to N players inside the general framework of symmetric multiplayer matrix games [13]. We fix $V = 4$, $C = 3$ for every numerical statement in this paper, and keep V and C symbolic wherever a result does not depend on their values.

One statistic drives every classical payoff: the Hawk count

$$k(x) = \sum_{j=1}^N x_j \in \{0, 1, \dots, N\}, \quad (1)$$

which counts how many players choose the aggressive action at outcome x . The payoff tensor then assigns to each player j and each outcome x the value

$$P_j(x) = \begin{cases} V/N, & k(x) = 0, \\ V/k(x), & 0 < k(x) < N \text{ and } x_j = 1, \\ 0, & 0 < k(x) < N \text{ and } x_j = 0, \\ (V - C)/N, & k(x) = N. \end{cases} \quad (2)$$

In words, the four branches of the payoff tensor, Eq. (2), say that when nobody fights the players split the asset evenly; when some but not all fight, the Hawks split the asset among themselves and the Doves receive nothing; and when everybody fights, the players split what the conflict leaves behind. We write $P(x) = (P_1(x), \dots, P_N(x))$ for the profile of per-outcome payoffs, and we reserve the term *payoff vector* for the vector of expected player payoffs introduced later in this section with the comparison and fairness metrics.

An $N = 4$ example makes the branches concrete. At the all-Dove outcome $x = 0000$ the Hawk count is $k = 0$, so every player receives $V/4$. At the one-Hawk outcome $x = 1000$ the Hawk count is $k = 1$: the lone Hawk receives $V/1 = V$ and each of the three Doves receives 0. At the all-Hawk outcome $x = 1111$ the Hawk count is $k = 4$, so every player receives $(V - C)/4$. Each branch keeps the player index, because P_j depends on the player’s own action x_j as well as on the aggregate $k(x)$: two players at the same outcome need not be paid the same. That per-player resolution is what later lets us measure how a quantum protocol redistributes value between graph positions rather than only how much value it creates in total.

Two properties of the payoff tensor, Eq. (2), carry through the rest of the paper. It is defined for every N and every one of the 2^N outcomes, so scaling the player count needs no special cases; and it depends on an outcome only through the pair $(k(x), x_j)$, which is what makes the welfare geometry of the next subsection tractable. Extending a two-player conflict game to N players under a shared entangling operation follows the multiplayer quantum-game construction of Benjamin and Hayden, who give worked $N = 3$ and $N = 4$ cases [6];

TABLE I
EVALUATION QUESTIONS, EVIDENCE LAYERS, AND INTERPRETATION BOUNDARIES.

Question	Operational test	Evidence	Interpretation permitted
Which phases support cooperation?	Compare every named unilateral strategy with the cooperative payoff.	Phase-boundary proof, unrestricted witness, and dense-circuit cross-checks.	Exact ideal statements for the specified operator and menu; no unrestricted equilibrium claim.
How does the entangler affect payoffs?	Evaluate the same fixed strategy and enumerate finite-menu best replies.	Five-family noiseless landscape, per-player vectors, and graph-symmetry checks.	A fixed-strategy comparison; not each family's globally optimal play.
What determines noise robustness?	Compare payoff gaps and equilibrium status; vary compilation and match gate counts.	Gate-level sweeps, implementation controls, and wiring permutations.	Production-circuit robustness; geometry and synthesis are not interchangeable.
What survives on hardware?	Test all-player deviation gaps under a fixed confidence rule; inspect welfare and population separately.	Preregistered circuits, raw counts, calibration records, and repeated batches.	Bounded implemented-game evidence; no distributed-market deployment claim.

Eq. (2) is the closed form for general N that this paper uses throughout. With the classical payoffs fixed, we can ask what they imply about equilibrium and total welfare.

B. Classical Equilibrium and Welfare Geometry

The payoff tensor says what each outcome pays, but not which outcome self-interested players reach, nor what the group holds in total once they get there; a quantum protocol can only be said to improve on the classical game once both are settled.

Hawk strictly dominates Dove at the parameters used here. Take player j facing m opponents who play Hawk, $0 \leq m \leq N - 1$, and read off the branches of the payoff tensor, Eq. (2), that player j can choose between. Playing Hawk raises the Hawk count to $m + 1$ and pays $V/(m + 1)$ when $m < N - 1$, or $(V - C)/N$ when $m = N - 1$. Playing Dove leaves the Hawk count at m and pays V/N when $m = 0$, or 0 otherwise. The three comparisons are V against V/N at $m = 0$; $V/(m + 1) > 0$ against 0 for $0 < m < N - 1$; and $(V - C)/N$ against 0 at $m = N - 1$. The first is strict for every $N \geq 2$, the second always, and the third exactly when $V > C$. At $V = 4$, $C = 3$ all three are strict, so Hawk pays more than Dove against every opponent configuration. The condition that produces dominance is $V > C$, not the Hawk–Dove structure itself, which is why we scope the statement to these parameters.

Strict dominance fixes the classical benchmark. Since Hawk is the better reply to every opponent configuration, the all-Hawk outcome $x = 1^N$ is the unique pure-strategy Nash equilibrium of the classical game [14]: no player gains by unilaterally switching to Dove, and no other profile survives one round of best replies. At $V = 4$, $C = 3$ it pays $(V - C)/N = 1/N$ per player. That $1/N$ is the classical baseline against which this paper measures every advantage.

All-Dove pays $V/N = 4/N$ per player and Pareto-dominates that equilibrium, so the classical game is a social dilemma in the standard sense: the strictly dominant action, taken by everyone, leaves everyone worse off than mutual cooperation [15]. At $N = 2$ the payoff tensor specializes to

the familiar two-player conflict matrix, paying $V/2$ to each player under mutual Dove, $(V, 0)$ under one-sided aggression, and $(V - C)/2$ to each under mutual Hawk. The interior mixed equilibrium usually associated with the Hawk–Dove game [12] is unavailable at these parameters. At $N = 2$, indifference between the two actions requires an opponent Hawk probability of V/C , and $V/C = 4/3$ falls outside the probability simplex whenever $V > C$; for general N , strict dominance excludes every interior mixed equilibrium. The all-Hawk pure equilibrium is therefore the only classical baseline this paper needs.

Individual payoffs move sharply between outcomes, but total value does not.

Proposition 1 (Outcome-level welfare). *For every $x \in \{0, 1\}^N$,*

$$\sum_{j=1}^N P_j(x) = V - C \mathbf{1}_{\{x=1^N\}}. \quad (3)$$

Proof. Split on the Hawk count $k(x)$. If $k(x) = 0$, every player is a Dove paid V/N and the N terms sum to V . If $0 < k(x) < N$, the $k(x)$ Hawks are each paid $V/k(x)$ and the $N - k(x)$ Doves are each paid 0, so the sum is $k(x) \cdot V/k(x) = V$ again. If $k(x) = N$, every player is a Hawk paid $(V - C)/N$ and the sum is $V - C$. The indicator selects that last case, the only one in which value is destroyed rather than redistributed. $\square$

The welfare identity, Eq. (3), controls mean payoff under any outcome distribution. Let p be the distribution over the 2^N outcomes produced by whatever procedure the players follow—classical randomization now, a quantum circuit from the next subsection onward—and let $\bar{\pi}$ be the resulting mean payoff per player, the average of the per-player expected payoffs defined with the metrics later in this section. Averaging the welfare identity over p gives

$$\bar{\pi} = \frac{1}{N} \sum_x p(x) \sum_{j=1}^N P_j(x) = \frac{V}{N} - \frac{C}{N} p(1^N). \quad (4)$$

In plain terms, mean payoff depends on the outcome distribution through one number: the probability of the all-Hawk outcome. Every other outcome, however unequal its per-player split, redistributes the same total value V . A protocol can therefore raise mean welfare only by suppressing $p(1^N)$, and it can do so while leaving individual players very unequally paid—a gap between mean and fairness that Sec. VI-A measures directly. What such a protocol looks like is the subject of the next subsection.

C. Quantum Strategies and the EWL Protocol

Classical play reaches all-Hawk and destroys C of the contested value, so the question is what changes when the players' actions are carried by qubits rather than by coins.

The Eisert–Wilkens–Lewenstein (EWL) protocol answers it in three steps [1]. One qubit is issued to each player and the register is entangled by a shared operation agreed before play. Each player then acts locally and privately on their own qubit, exactly as a classical player acts privately on their own choice. Finally the entangling operation is undone and the register is measured, which returns one of the 2^N classical outcomes of Sec. IV-A and pays the players by the payoff tensor, Eq. (2). The players never share a signal during play; all correlation is set up in advance and released by the measurement.

A player's action is a single-qubit unitary. We take the full special unitary group, writing a strategy in the three-parameter form

$$U(\theta, \alpha, \beta) = \begin{pmatrix} e^{i\alpha} \cos(\theta/2) & ie^{i\beta} \sin(\theta/2) \\ ie^{-i\beta} \sin(\theta/2) & e^{-i\alpha} \cos(\theta/2) \end{pmatrix}, \quad (5)$$

with $\theta \in [0, \pi]$ and $\alpha, \beta \in [-\pi, \pi]$. The angle θ controls how much a player rotates between the two classical actions, while α and β set phases that have no classical counterpart and are the whole source of the protocol's extra structure.

Three named strategies recur throughout the paper:

$$\begin{aligned} D &= U(0, 0, 0), & H &= U(\pi, 0, 0), \\ Q_N &= U(0, \pi/N, \pi/N). \end{aligned} \quad (6)$$

Here D is the identity and leaves a player's qubit in the Dove state, H equals $i\sigma_x$ and flips it to Hawk (the leading factor i is a global phase and is unobservable), and Q_N is the phase-only strategy $\text{diag}(e^{i\pi/N}, e^{-i\pi/N})$, which has no classical counterpart at all. The π/N scaling makes Q_N GHZ-derived: it is one choice of phase that gives the GHZ entangler's two branches a common sign (Sec. IV-G), so the familiar two-player $Q = U(0, \pi/2, \pi/2)$ is the $N = 2$ case of one family rather than a universal quantum strategy. We hold Q_N fixed when we change entangler family, so that a family comparison changes the correlation structure and nothing else.

Writing U_j for the strategy of player j and $J_T(\gamma)$ for the entangling operation of family T at entanglement angle γ , the

protocol produces

$$|\psi_{\text{out}}\rangle = J_T^\dagger(\gamma) \left(\bigotimes_{j=1}^N U_j \right) J_T(\gamma) |0^N\rangle, \quad (7)$$

and measuring in the computational basis gives each classical outcome the probability

$$p_T(x | U) = |\langle x | \psi_{\text{out}} \rangle|^2, \quad (8)$$

where x_j is the measured value of player j 's qubit, matching the bitstring convention of Sec. IV-A. The two equations say that the protocol is a map from a strategy profile to a distribution over the same classical outcomes the classical game had; the payoff rule is untouched, and every result in this paper comes from moving probability between those outcomes. The entanglement angle γ controls the interpolation used to correlate the register before the players act: every family we study satisfies $J_T(0) = I$, the identity on all N qubits, so $\gamma = 0$ returns the classical game exactly. We use $\gamma = \pi/2$ as the largest nominal interpolation angle in the study; whether that endpoint is maximally entangled depends on the family and entanglement measure and is not assumed uniformly across the five families. Chappell et al. use the symbol θ for the corresponding GHZ-family angle [7]. The subscript T is the only thing left undefined here; the five entangler families it ranges over are the subject of the next subsection.

One scope limit must be stated before any equilibrium claim. Players physically hold the whole strategy space of Eq. (5), but the equilibrium tests in this paper enumerate deviations over the restricted menu $\{D, H, Q_N\}$ only. That restriction is not cosmetic: Benjamin and Hayden showed that widening the deviation space to all of $\text{SU}(2)$ removes the equilibrium EWL originally claimed, because an ideal counter-strategy always exists [4]. Every equilibrium statement below is therefore a restricted-menu equilibrium, and we never call it a full- $\text{SU}(2)$ result. We also work in the unitary EWL quantization rather than the alternative probabilistic scheme of Marinatto and Weber [16], which is formulated for two players only.

D. Entangler Families and Graph Symmetry

The output state, Eq. (7), is only as specific as the entangler $J_T(\gamma)$ inside it, so the five families that T ranges over must be written down before any result can be attributed to correlation structure rather than to strategy.

We compare five entangler families: GHZ, W, ring, star, and fully connected. They are not all the same kind of object, and the paper keeps two levels of vocabulary apart because of it. Ring, star, and fully connected are *graph topologies*: each is fixed by an edge set $E(G)$ on the players, and its entangler is a product of one two-qubit rotation per edge. GHZ and W carry no edge set at all. Each is a single N -body operator interpolating from the identity toward one designated entangled state, so hub, leaf, and graph-position language is

undefined for them. We therefore write “entangler family” for any five-way statement and reserve “graph topology” for the pairwise three. Both global constructions are included because at three qubits GHZ and W are the two classes of genuine tripartite entanglement that stochastic local operations cannot interconvert [17], and because comparing GHZ-type against W-type correlation in N -player quantum games is an established question [7].

The GHZ family entangles the whole register with a single global rotation. We define

$$J_{\text{GHZ}}(\gamma) = \cos \frac{\gamma}{2} I^{\otimes N} + i \sin \frac{\gamma}{2} X^{\otimes N}, \quad (9)$$

where X is the Pauli bit flip and I the single-qubit identity. Since $X^{\otimes N}$ flips every bit at once, Eq. (9) couples each classical outcome only to its bitwise complement, and in particular couples all-Dove to all-Hawk: applied to $|0^N\rangle$ at $\gamma = \pi/2$ it prepares $(|0^N\rangle + i|1^N\rangle)/\sqrt{2}$. That maximally entangled member, $(I^{\otimes N} + iX^{\otimes N})/\sqrt{2}$, is the entangling gate Benjamin and Hayden use for multiplayer quantum games [6]; Eq. (9) is the one-parameter interpolation of it that lets γ act as a control variable.

The graph topologies replace that one global rotation with one rotation per edge. For a graph G on the players we define

$$J_G(\gamma) = \prod_{(r,s) \in E(G)} \exp\left(i \frac{\gamma}{2} X_r X_s\right), \quad (10)$$

where X_r flips player r ’s qubit alone. All the operators $X_r X_s$ commute, so the product does not depend on the order in which the edges are taken and equals one exponential of their sum. The three edge sets are the cycle C_N , with $E = \{(j, j+1) : 1 \leq j < N\} \cup \{(N, 1)\}$; the star S_N , with $E = \{(1, j) : 2 \leq j \leq N\}$, whose distinguished player 1 is the hub and whose remaining $N - 1$ players are the leaves; and the complete graph K_N , with every pair, which defines the fully connected family. We index players $1, \dots, N$ as in Sec. IV-A, while the implementation indexes qubits from zero and pins the hub at qubit 0, which is player 1 here.

Writing $m_G = |E(G)|$ for the edge count, the standard formulas are scoped to simple graphs with at least three vertices. For $N \geq 3$, the standard counts are $m_{C_N} = N$, $m_{S_N} = N - 1$, and $m_{K_N} = N(N - 1)/2$. At $N = 2$, C_2 degenerates to the single edge $(1, 2)$, so $m_{C_2} = 1$; the star and complete graph are the same single edge. Thus all three graph topologies are one operator and the distinction between them is empty. At $N = 3$ the cycle C_3 is a triangle, which is K_3 , so ring and fully connected are again the same operator and every ring-versus-fully-connected statement in this paper is a claim about $N \geq 4$. The star is distinct from both at every $N \geq 3$ and is the only graph topology whose players are not all structurally interchangeable. Because each edge is realized by one two-qubit rotation, m_G is also the graph entangler’s two-qubit gate count before routing; the counts actually executed on hardware are reported in Sec. VI-B.

The W family interpolates toward a different target state. Let $|W\rangle = N^{-1/2} \sum_{j=1}^N |e_j\rangle$, where e_j is the outcome in which player j alone plays Hawk, and let S_W be the reflection that exchanges $|0^N\rangle$ with $|W\rangle$:

$$S_W = I^{\otimes N} - |0^N\rangle\langle 0^N| - |W\rangle\langle W| + |0^N\rangle\langle W| + |W\rangle\langle 0^N|. \quad (11)$$

In words, S_W swaps the all-Dove outcome with the equal superposition of the N one-Hawk outcomes and acts as the identity on everything orthogonal to both.

Because S_W is Hermitian and squares to the identity, the interpolation that produced Eq. (9) applies to it unchanged:

$$J_W(\gamma) = \cos \frac{\gamma}{2} I^{\otimes N} + i \sin \frac{\gamma}{2} S_W, \quad (12)$$

which equals $\exp(i\gamma S_W/2)$, is unitary at every γ , and satisfies $J_W(0) = I$ like the other four families. On the all-Dove input, the state at angle γ is $J_W(\gamma)|0^N\rangle = \cos(\gamma/2)|0^N\rangle + i \sin(\gamma/2)|W\rangle$; hence at $\gamma = \pi/2$ the prepared state is $(|0^N\rangle + i|W\rangle)/\sqrt{2}$, not $|W\rangle$ alone. We call $|W\rangle$ the exchanged target component. This interpolation is a construction we choose: that component is fixed by the family’s name, but the generator’s action beyond the $\text{span}\{|0^N\rangle, |W\rangle\}$ branch is not. We adopt Eq. (11) so that GHZ and W share the same cosine–sine interpolation law on the initial-state branch. Their generators remain different operators away from that branch. Every W result reported here is therefore a statement about this construction, not about W-type entanglement in general.

Equal γ across families does not mean equal interaction strength, and a family comparison has to say so. Eq. (9) and Eq. (12) apply one rotation through γ , while Eq. (10) applies m_G of them, so at the same nominal angle the complete graph carries $N(N - 1)/2$ rotations against the star’s $N - 1$. We define the normalized interaction-strength convention

$$J_G^n(\gamma) = \prod_{(r,s) \in E(G)} \exp\left(i \frac{\gamma}{2m_G} X_r X_s\right), \quad (13)$$

which spreads a single unit of rotation angle γ evenly over the edges so that every family carries the same summed generator weight. Two limits of the convention belong with the definition: it equalizes total rotation angle, not entangling power or gate count, and it leaves Eqs. (9) and (12) untouched, since a single global rotation already has $m = 1$.

Every reported entangler-family comparison is computed under the unnormalized convention, with all five families evaluated at the same raw γ . We define Eq. (13) here so that the family comparison of Sec. VI-A can state exactly what it does not control for: a star-versus-fully-connected difference at equal raw γ confounds correlation structure with coupling budget, and separating the two requires re-running the comparison under Eq. (13). That sensitivity control is not among the results reported here.

Relabeling the players acts on the graph topologies in the obvious way, and that is what makes graph position a well-defined variable rather than an artifact of naming. For a

permutation σ of the players let R_σ be the unitary that moves the contents of qubit r to qubit $\sigma(r)$, so that $R_\sigma X_r R_\sigma^\dagger = X_{\sigma(r)}$ and $R_\sigma |x\rangle = |\sigma(x)\rangle$ with $\sigma(x)_{\sigma(j)} = x_j$. Conjugating Eq. (10) factor by factor carries the edge (r, s) to $(\sigma(r), \sigma(s))$ and leaves the angle alone, so

$$R_\sigma J_G(\gamma) R_\sigma^\dagger = J_{\sigma(G)}(\gamma), \quad (14)$$

where $\sigma(G)$ has edge set $\{(\sigma(r), \sigma(s)) : (r, s) \in E(G)\}$. In words, entangling and then relabeling is the same as relabeling and then entangling on the relabeled graph. The global families are the degenerate case: $X^{\otimes N}$, $|0^N\rangle$, and $|W\rangle$ are each invariant under every permutation, so $R_\sigma J_{\text{GHZ}} R_\sigma^\dagger = J_{\text{GHZ}}$ and $R_\sigma J_W R_\sigma^\dagger = J_W$ for all σ .

Covariance of the operator becomes covariance of expected payoffs once the players act symmetrically. We state the result directly in terms of outcome probabilities and the payoff tensor, leaving the payoff-vector notation to the next subsection.

Proposition 2 (Relabeling covariance of payoffs). *Let every player use the same strategy U , let G be a graph topology, and let any noise acting on the register be a channel that commutes with R_σ . Then for every permutation σ and every player j ,*

$$\sum_x p_{\sigma(G)}(x) P_{\sigma(j)}(x) = \sum_x p_G(x) P_j(x). \quad (15)$$

In particular, if σ is an automorphism of G , then any two players in the same automorphism orbit receive equal expected payoffs.

Proof. Three facts drive the argument: the all-Dove state is fixed by relabeling, $R_\sigma |0^N\rangle = |0^N\rangle$; a symmetric strategy layer commutes with relabeling, $R_\sigma \bigotimes_j U = \bigotimes_j U R_\sigma$; and the entangler transforms by Eq. (14). Substituting the three into the output state, Eq. (7), gives $|\psi_{\text{out}}(\sigma(G))\rangle = R_\sigma |\psi_{\text{out}}(G)\rangle$, so the outcome probabilities of Eq. (8) satisfy $p_{\sigma(G)}(\sigma(x)) = p_G(x)$. The payoff tensor, Eq. (2), depends on an outcome only through the Hawk count and the player's own bit; relabeling preserves the Hawk count, $k(\sigma(x)) = k(x)$, and carries x_j to $\sigma(x)_{\sigma(j)}$, so $P_{\sigma(j)}(\sigma(x)) = P_j(x)$. Multiplying these identities, summing over all 2^N outcomes, and changing the dummy variable from $\sigma(x)$ to x gives Eq. (15). A noise channel that commutes with R_σ leaves every step valid at the density-matrix level. The orbit claim is the case $\sigma(G) = G$. $\square$

The orbit structure of each family therefore fixes which payoff equalities must hold before any circuit is run. The ring and the fully connected graph are vertex-transitive, so their players form a single orbit and all N expected payoffs coincide. The star has exactly two orbits, the hub alone and the $N - 1$ leaves: the leaves must be paid equally, while the hub is free to differ. GHZ and W are invariant under the full symmetric group, so their players also form one orbit. Reading structural symmetry off a graph, and keeping it distinct from the equivalences of the state that graph defines, is standard for graph-defined multiparty entanglement [18]. The proposition

says nothing about how far apart two different orbits sit, which is a quantitative question; Sec. VI-A measures the hub-leaf separation, and the next subsection defines the metrics in which it is expressed.

E. Expected Payoff, Advantage, Equilibrium, and Fairness

The outcome distribution becomes economically meaningful only after we map it to player payoffs. For strategy profile U and entangler family T , we define

$$\pi_j(U; T) = \sum_x p_T(x | U) P_j(x), \quad \pi = (\pi_1, \dots, \pi_N). \quad (16)$$

Thus π_j averages player j 's payoff tensor over all measured outcomes, and π records the full player-level allocation. The mean payoff introduced in Eq. (4) therefore specializes to

$$\bar{\pi}(U; T) = \frac{1}{N} \sum_{j=1}^N \pi_j(U; T). \quad (17)$$

The payoff vector remains the primary evidence; the mean summarizes total performance per player but cannot show who gains or loses.

Incentive compatibility requires player-level deviation checks rather than a high cooperative payoff. For a fixed entangler family T , and a fixed implementation and noise path where applicable, we suppress the T argument. With $Q_{N,-j}$ denoting the other $N - 1$ players at Q_N , we define the unilateral-deviation gap

$$g_j(a) = \pi_j(Q_N^{\otimes N}) - \pi_j(a, Q_{N,-j}), \quad a \in \{D, H\}. \quad (18)$$

The profile $Q_N^{\otimes N}$ passes the restricted-menu equilibrium test if and only if $g_j(a) \geq 0$ for every player j and both deviations $a \in \{D, H\}$. This is a test within $\{D, H, Q_N\}$, not against unrestricted $\text{SU}(2)$; deviating to Q_N itself leaves the profile unchanged.

Two payoff comparisons answer different questions and therefore receive separate names and symbols. We define the analytic-baseline advantage and the circuit-relative payoff gap by

$$\Delta_{\text{ana}}(U; T) = \bar{\pi}(U; T) - \frac{V - C}{N}, \quad (19)$$

and

$$\Delta_{\text{circ}}(T) = \bar{\pi}(Q_N^{\otimes N}; T) - \max_{a \in \mathcal{E}_T^{D,H}} \bar{\pi}(a; T), \quad (20)$$

where $\mathcal{E}_T^{D,H}$ is the set of pure restricted $\{D, H\}^N$ circuit equilibria evaluated under the same entangler family and, where applicable, the same implementation and noise path. The first equation compares with the fixed analytic all-Hawk baseline; the second compares with a circuit-relative classical equilibrium.

Metric	Comparator
analytic-baseline advantage	Fixed analytic baseline $(V - C)/N$
circuit-relative payoff gap	Highest-mean pure restricted $\{D, H\}^N$ circuit equilibrium under the same family and path

Their values and zero crossings are not interchangeable, so we never use a bare “advantage” label when the comparator could be unclear.

State concentration is recorded separately from economic payoff. We define the all-zero target-state population by

$$P(0^N) \equiv p_T(0^N | U). \quad (21)$$

This target-state population is the probability of one computational-basis outcome. For the explicit pure target $|0^N\rangle$ it equals the squared state fidelity, $F(\rho, |0^N\rangle) = \langle 0^N | \rho | 0^N \rangle$. It is not fidelity to an arbitrary ideal entangled state and does not determine the remaining populations or coherences. It is distinct from the payoff entry $P_j(x)$.

Finally, we summarize player-level dispersion only after reporting the vector. The fairness range and player floor are

$$S_\pi = \max_j \pi_j - \min_j \pi_j, \quad F_\pi = \min_j \pi_j. \quad (22)$$

Here $S_\pi = 0$ means equal expected payoffs and F_π identifies the worst-paid player. These scalar fairness summaries do not replace the payoff vector, which remains the primary fairness evidence. The next subsection applies these metrics to the exact GHZ cooperative benchmark.

F. GHZ Cooperative Benchmark

We now derive the cooperative reference against which unilateral deviations are measured. For the GHZ entangler, the phase structure of Q_N makes this benchmark exact at every entanglement angle, not only at maximal entanglement.

Proposition 3 (Cooperative invariance in the entanglement angle). *For every γ ,*

$$J_{\text{GHZ}}^\dagger(\gamma) Q_N^{\otimes N} J_{\text{GHZ}}(\gamma) |0^N\rangle = -|0^N\rangle, \quad (23)$$

hence $p(0^N) = 1$, $\pi_j = V/N$, and $\Delta_{\text{ana}} = C/N$ independently of γ .

The minus sign is a global phase, so measurement returns the all-Dove outcome with certainty for any entanglement angle.

Proof. Using the GHZ operator of Eq. (9) and the phase gate Q_N of Eq. (6), we obtain

$$\begin{aligned} J_{\text{GHZ}}(\gamma) |0^N\rangle &= \cos\left(\frac{\gamma}{2}\right) |0^N\rangle \\ &\quad + i \sin\left(\frac{\gamma}{2}\right) |1^N\rangle, \\ e^{iN\pi/N} &= e^{i\pi} = -1, \quad e^{-iN\pi/N} = e^{-i\pi} = -1, \\ Q_N^{\otimes N} |0^N\rangle &= -|0^N\rangle, \quad Q_N^{\otimes N} |1^N\rangle = -|1^N\rangle, \\ Q_N^{\otimes N} J_{\text{GHZ}}(\gamma) |0^N\rangle &= -J_{\text{GHZ}}(\gamma) |0^N\rangle, \\ J_{\text{GHZ}}^\dagger(\gamma) [-J_{\text{GHZ}}(\gamma) |0^N\rangle] &= -|0^N\rangle. \end{aligned}$$

The phase equalities show that both GHZ branches acquire exactly the same phase -1 ; the final steps show that disentangling recovers $-|0^N\rangle$. $\square$

The payoff identities follow directly from the outcome-level identities above. Since $p(0^N) = 1$, Eq. (2) gives $\pi_j = V/N$ for every player, and substituting the resulting mean payoff into Eq. (19) gives $\Delta_{\text{ana}} = V/N - (V - C)/N = C/N$.

Corollary ($V = 4$, $C = 3$). At the maximally entangled experimental benchmark, $\gamma = \pi/2$, with $V = 4$ and $C = 3$,

$$\pi_j = \frac{4}{N}, \quad \Delta_{\text{ana}} = \frac{3}{N}. \quad (24)$$

Thus every player receives $4/N$, exactly $3/N$ above the analytic all-Hawk baseline.

For the cooperative benchmark, γ is therefore a pure incentive dial: it can move the deviation incentives characterized in Sec. IV-G while leaving the cooperative payoff exactly unchanged.

G. Incentive-Compatibility Boundary

The cooperative output does not determine whether a player benefits from departing from it. We characterize the original phase choice and then introduce a distinct branch that changes its restricted incentives.

Let $Q_{N,m} = U(0, \alpha_m, \alpha_m)$, where $\alpha_m = m\pi/N$ and m is an integer. The original strategy Q_N is $Q_{N,1}$. Both GHZ branches acquire the common phase $(-1)^m$, so the proof in Sec. IV-F gives cooperative payoff V/N for every branch and every γ . The identity branch $m = 0$ also cooperates in outcome; that observation alone establishes no incentive compatibility.

Proposition 4 (Restricted phase-branch incentives). *For $N \geq 2$, $V > 0$, $C > 0$, and $t = \sin^2 \gamma \sin^2 \alpha_m$, the unilateral Hawk and Dove payoffs against $Q_{N,m}$ are*

$$\begin{aligned} \pi_j(H, Q_{N,m,-j}) &= V(1 - t), \\ \pi_j(D, Q_{N,m,-j}) &= (V - Ct)/N. \end{aligned} \quad (25)$$

The cooperative profile is a Nash equilibrium within $\{D, H, Q_{N,m}\}$ if and only if

$$\sin^2 \gamma \sin^2 \alpha_m \geq 1 - \frac{1}{N}. \quad (26)$$

The inequality requires sufficient entanglement and an appropriate cooperative phase; it is not an equilibrium condition for arbitrary $\text{SU}(2)$ operations.

Proof. The other players contribute total phase $(N-1)\alpha_m = m\pi - \alpha_m$. For a Hawk deviation, applying $J^\dagger$ leaves support on the deviator-only Hawk outcome e_j and its complement. The probability of e_j is $|\cos^2(\gamma/2)e^{-i\alpha_m} + \sin^2(\gamma/2)e^{i\alpha_m}|^2 = 1 - t$. The deviator earns V on e_j and zero on its complement. A Dove deviation leaves support on 0^N and 1^N , with probabilities $1 - t$ and t , respectively, giving $(V - Ct)/N$. Comparing each payoff with V/N proves the condition, since Dove never benefits and Hawk does not benefit exactly when $t \geq 1 - 1/N$. $\square$

For the original branch at $\gamma = \pi/2$, the Hawk payoff is $V \cos^2(\pi/N)$, equal to $2 + 2 \cos(2\pi/N)$ at $V = 4$. It exceeds V/N for $N \geq 4$, matching the fixed-profile simulations of Sec. VI-A. Define a separate alternative strategy

$$Q_N^* = Q_{N, \lfloor N/2 \rfloor}. \quad (27)$$

This selects the largest branch phase α_m not exceeding $\pi/2$. For $N = 2, 3$, $\lfloor N/2 \rfloor = 1$, so $Q_N^* = Q_N$; the two branches differ only for $N \geq 4$. For even N the Hawk payoff is zero at maximal entanglement; for odd $N \geq 3$ it is $V \sin^2(\pi/(2N)) < V/N$, since $\sin^2(\pi/(2N)) < \pi^2/(4N^2) < 1/N$. The Dove inequality is also strict. Thus Q_N^* is a strict $\{D, H, Q_N^*\}$ equilibrium for every $N \geq 2$ in the ideal maximally entangled game. Mixtures over the same menu cannot improve on its best pure deviation. The old and alternative menus are explicitly different; no original-phase simulation or hardware result is relabeled.

The unrestricted boundary is explicit. Against $Q_{N,m}$, $U(\pi, 0, -\alpha_m)$ yields the sole-Hawk outcome e_j with certainty: the two amplitudes contributing to e_j have equal phase and their weights sum to $\cos^2(\gamma/2) + \sin^2(\gamma/2) = 1$. The payoff is V , a gain of $V(1 - 1/N)$, for every γ . Neither cooperative branch is therefore an equilibrium against all $SU(2)$ deviations. This is a profile-specific witness, not a nonexistence theorem for other pure or mixed equilibria; the importance of specifying the strategy space is established in [4].

Ideal existence also differs from uniform robustness. For even N , Eq. (26) gives $\gamma_c = \arccos(1/\sqrt{N})$ on $[0, \pi/2]$, so the permitted angle window narrows as N grows, while absolute incentive margins decrease. The simulation section evaluates the alternative branch separately from the original fixed-strategy landscape.

V. COMPUTATION AND VALIDATION PROTOCOL

A. Exact simulation and strategy evaluation

Noiseless experiments use the dense EWL operator and exact statevector probabilities, with player j mapped to bit $j - 1$ of the integer outcome. The restricted equilibrium search enumerates all 3^N profiles in the named menu and checks every unilateral change. Symmetric continuous profile searches are candidate-generation procedures. A fixed-opponent $SU(2)$ best response is evaluated using the largest eigenvalue of a four-dimensional quadratic form, following the methodological precedent in [19]. Every player position must pass before a candidate is certified; a hub-only check is insufficient for a star.

B. Gate-level noise and compilation controls

The noisy path reconstructs each dense entangler from elementary gates, transpiles to the pinned $\{u, cx\}$ basis, and applies depolarizing channels after those gates. For a d -dimensional gate subsystem the channel is $\mathcal{D}_p(\rho) = (1 - p)\rho + pI_d/d$ when acting on that subsystem's density operator;

on a larger correlated register it replaces the subsystem with I_d/d and retains the complementary reduced state. The main sweep uses equal nominal strengths on one- and two-qubit gates. The gate-level and dense operators agree up to global phase, and the $p = 0$ density-matrix output is checked against the noiseless path.

Entangler synthesis is cached by family, player count and angle. It is independent of the selected strategy profile. The reported noise ranking therefore refers to the specified decomposition and compilation settings. Matched-count and varied-compilation controls test that dependency rather than treating a circuit's gate count as an intrinsic topology invariant.

C. Uncertainty and preregistered decisions

Raw payoff estimates are linear functions of outcome frequencies; their multinomial covariance captures shot noise within the sampled circuit distribution. Before every hardware submission, the predictions, pass criteria and qubit assignment were written down and committed to the code repository; we call these tests preregistered and report failures alongside passes. The alternative-phase all-player hardware tests use simultaneous one-sided Hoeffding bounds over the full deviation family. The exact rule and its observed outcomes appear in Sec. VI-B5. Readout correction and zero-noise extrapolation in the original-phase experiments use their separately recorded protocols. Neither shot-noise intervals nor within-session replication establish stability across independent calibration epochs.

D. Experimental Setup and Safeguards

Every submission passes three gates: (i) an Operator.equiv circuit-identity assertion against the dense $J^\dagger(U^{\otimes N})J$ reference, (ii) a noiseless simulator dry-run that must reproduce the exact analytic advantage, and (iii) a full rehearsal of the batch, including mitigation analysis, on the device noise model reduced to the executed qubits. The execution configuration of each run family is summarized in Table II.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

A. Simulation Landscape

1) *Scaling and Topology at Zero Noise:* For GHZ, the fixed profile $(Q_N, \dots, Q_N)$ pays exactly $V/N = 4/N$ per player; its circuit-evaluated restricted $\{D, H\}$ comparator is $1/N$, giving a circuit-relative payoff gap of $3/N$ (Figs. 1 and 2). The other topologies are evaluated cell by cell. Fully connected attains V/N for all recorded $N = 2-6$ cells; ring does so except at $N = 4$; W attains V/N but has a larger circuit comparator and therefore a smaller gap; and star attains V/N through $N = 5$ but deviates at $N = 6$, with asymmetric player payoffs from $N = 4$ onward. The restricted-menu equilibrium status is likewise cell-specific: GHZ passes the $\{D, H, Q_N\}$ deviation check at $N = 2, 3$, and ring and fully connected also pass at $N = 3$.

TABLE II
HARDWARE EXECUTION CONFIGURATION

Parameter	Value
Device	ibm_fez (IBM Heron r2)
Shots per circuit	4096
$N = 3$ validation chain	[136, 143, 142]
Scaling chain ($N = 3, 4, 5$)	[59, 75, 74, 73, 79]
Extension chain ($N = 3 \dots 7$)	[97, 107, 108, 109, 110, 111, 98]
Alternative-phase chain	[141, 142, ..., 147]
Readout mitigation	tensored confusion-matrix inversion (original-phase runs only)
Zero-noise extrapolation	local CZ folding, $\lambda \in \{1, 3, 5\}$
Transpilation	optimization level 3, pinned layout (level 1 for alternative-phase runs)
Circuits per batch	11 scaling, 17 extension, 49 multi-topology, 70 alternative-phase
Software (original phase)	qiskit 1.3.2 / qiskit-aer 0.14.2 / qiskit-ibm-runtime 0.36.1
Software (alternative phase)	qiskit 2.5.2 / qiskit-aer 0.17.2 / qiskit-ibm-runtime 0.49.0

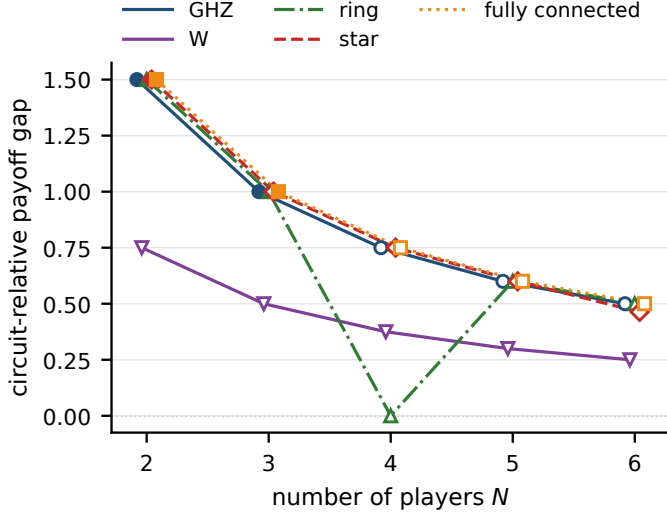


Fig. 1. Zero-noise circuit-relative payoff gap of the fixed profile $(Q_N, \dots, Q_N)$, defined as its mean payoff minus the highest-mean circuit-evaluated restricted $\{D, H\}$ equilibrium payoff, at $\gamma = \pi/2$ and $V = 4, C = 3$. The $3/N$ gap is attained by GHZ and fully connected at every recorded N ; the ring has zero gap at $N = 4$; the star shows a weaker $N = 6$ value (0.463 vs. 0.5); and W tracks below throughout. Filled markers indicate that the profile passes the restricted $\{D, H, Q_N\}$ deviation check, which depends on topology.

For GHZ, Eq. (25) gives the unilateral Hawk payoff $V \cos^2(\pi/N) = 2 + 2 \cos(2\pi/N)$, and the enumerated restricted deviation check agrees with $2 + 2 \cos(2\pi/N) \leq V/N$ for every $N = 2-8$. Thus at $N = 4$ the deviation pays 2.0 against the cooperative 1.0, and at $N = 5$ it pays $\varphi^2 \approx 2.618$ against 0.8. For $N \geq 4$ we therefore report the GHZ profile as a *fixed cooperative protocol*, not as an equilibrium prediction. The landscape also contains an $N = 4$ ring interference zero and a star $N = 6$ deviation from V/N ; for star the circuit-relative payoff gap is asymmetric across players: at $N = 4$ the three leaves gain 1.0 each while the hub gains nothing.

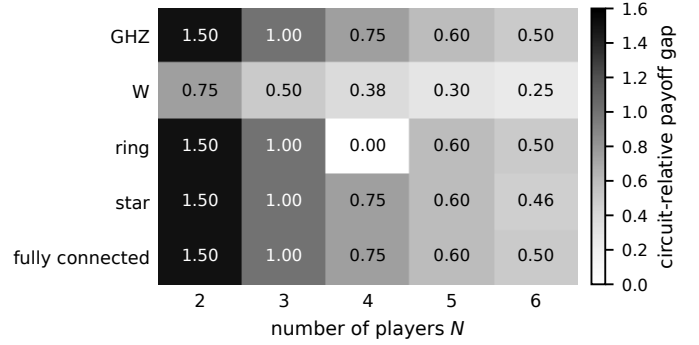


Fig. 2. Circuit-relative payoff-gap heatmap across the $(N, \text{topology})$ grid at $V = 4, C = 3$ (mean per-player payoff of $(Q_N, \dots, Q_N)$ minus the highest-mean circuit-evaluated restricted $\{D, H\}$ equilibrium payoff). GHZ and fully connected hold the $3/N$ gap everywhere; the only zero cell is the ring at $N = 4$.

2) *Entanglement-Angle Dependence*: Sweeping γ separates a cooperative payoff gap from an entanglement-enabled restricted-menu equilibrium. At $\gamma = 0, J = I$ and the phase-only Q_N profile is measurement-equivalent to all-Dove: its positive payoff gap over all-Hawk is classical in outcome probabilities and the profile does not pass the restricted deviation check. At nonzero γ , the same sweep records when the fixed profile begins to pass that finite-menu check.

At $N = 2$ every topology's circuit-relative payoff gap is positive from $\gamma = 0$, and (Q, Q) begins to pass the restricted deviation check from $\gamma = 0.3\pi$, the first grid point above the analytic GHZ threshold $\gamma_c = \pi/4$ of Eq. (26) (W excepted, which never passes in range). At $N = 3$ GHZ, fully connected, and ring hold a gap of 1.0 across the sweep and begin to pass at $\gamma = 0.3\pi-0.4\pi$; GHZ first passes at 0.4π , consistent with its analytic threshold $\arcsin \sqrt{8/9} \approx 0.392\pi$. Star and W never pass. At $N = 4$ (Fig. 3) GHZ and star hold 0.75 at every γ , fully connected dips to 0.40 mid-sweep and recovers, W decays monotonically to 0.375, and ring reaches exactly zero at $\gamma = \pi/2$, the same $N = 4$ interference zero. No series has a negative circuit-relative payoff gap; only that ring endpoint reaches zero.

3) *Player Symmetry Breaking in the Star Topology*: The star is the only topology that breaks player symmetry at zero noise. For $N \geq 4$ its per-player circuit-relative payoff gaps are uneven (Fig. 4): at $N = 4$ the three leaves have gap 1.0 each while the hub has 0.0; by $N = 6$ the hub has 0.952 against the leaves' 0.366. The mean circuit-relative payoff gap stays close to $3/N$ until $N = 6$, where the hub-leaf split pulls it to 0.463.

4) *Noise Robustness: Two Criteria Disagree*: Sweeping the depolarizing strength p from 0 to 0.05 on the gate-level noisy path, two natural "robustness" criteria give different orderings, so we state which we mean.

a) *Restricted-menu equilibrium criterion*: Under this criterion, the GHZ cooperative profile passes at $p = 0$ only for $N = 2, 3$, and at $N = 3$ it loses that status between $p = 0.04$

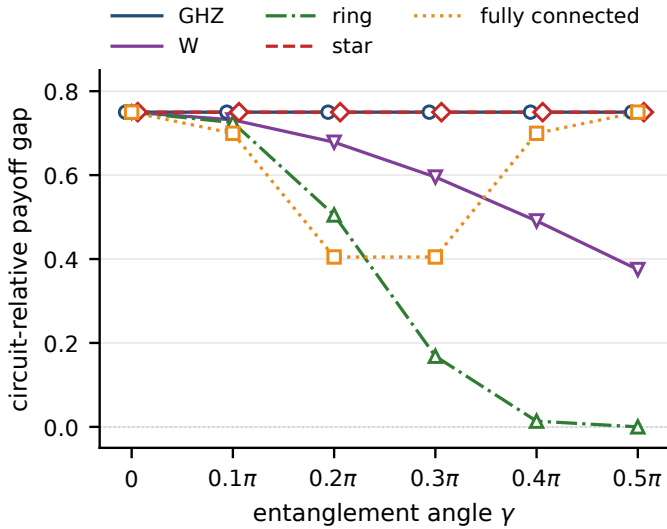


Fig. 3. Circuit-relative payoff gap versus entanglement angle γ at $N = 4$ ($V = 4, C = 3$, noiseless). GHZ and star hold 0.75 flat; fully connected dips and recovers; W decays monotonically; and ring falls to zero at $\gamma = \pi/2$. Filled markers indicate a passed restricted deviation check.

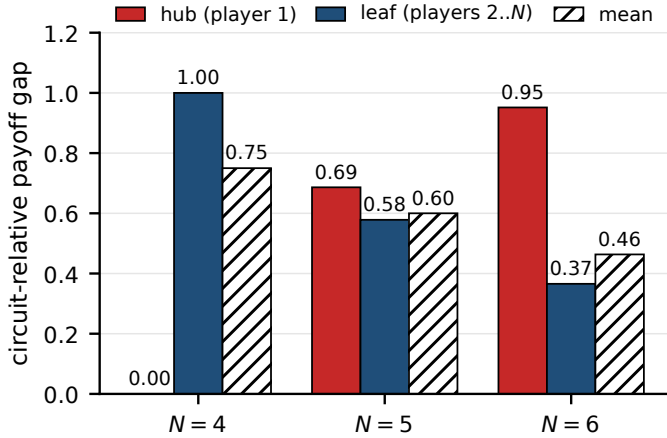


Fig. 4. Star-topology per-player circuit-relative payoff gap (hub, player 1, vs. each leaf) at $\gamma = \pi/2$, $V = 4, C = 3$, noiseless, for $N = 4 \dots 6$. Hatched bars show the mean over players; the hub gap at $N = 4$ is zero.

and 0.045. For $N = 4, 5$ the GHZ thresholds $p^* \approx 0.0197$ and 0.0098 are *circuit-relative payoff-gap zero crossings*, not equilibrium losses: the profile does not pass the restricted deviation check there even at $p = 0$. At zero noise, ring passes at $N = 2, 3$, fully connected at $N = 2, 3$, star at $N = 2$, and W at neither $N = 2$ nor $N = 3$.

b) Circuit-relative payoff-gap magnitude criterion: Under this criterion the picture flips: at $N = 3$ GHZ retains the largest mean gap at $p = 0.05$ (0.333, 33% of its noiseless value) versus ring 0.200 and W 0.047. These are not crossings relative to the analytic $1/N$ baseline: noise can change both the fixed-profile payoff and the highest-mean restricted $\{D, H\}$ circuit equilibrium, and comparator drift can drive a zero crossing. The natural-budget ordering therefore characterizes

TABLE III
INITIAL ADVANTAGE DECAY RATE PER TWO-QUBIT GATE, λ/CX WITH $\lambda = -(dA/dp)/A(0)$ ESTIMATED AT $p = 0.005$, UNDER GATE-LEVEL DEPOLARIZING NOISE. RING $N = 4$ HAS $A(0) = 0$.

N	GHZ	W	ring	star	fully conn.
3	2.55	0.98	2.44	2.65	2.44
4	2.25	0.79	—	2.42	2.10
5	2.09	0.79	2.13	2.25	1.82

the production circuits. Matched-count controls reduce gate-budget confounding within the XX family, whose normalized per-CX decay rates are similar at approximately 1.8–2.7 (Table III); no statistical equivalence test was recorded. Absolute GHZ–W payoff-gap ordering can remain after those controls because their zero-noise gaps differ. W is the normalized-retention outlier at approximately 0.8–1.0 per CX. It keeps a positive but small mean circuit-relative gap across the whole grid, while the mean hides per-player losses: at $N = 5$ two of the five players have negative per-player gaps from $p = 0.01$ (worst -0.077 at $p = 0.015$). All entanglers here are exact gate-level circuits; W uses an exact gate-level construction (44 and 68 CX per J at $N = 4, 5$, versus 100/444 for the retired transpiled-unitary fallback).

5) *Position-Locked Unfairness:* The noise-induced per-player asymmetry is not a property of the player labels or the ideal state. A wiring-permutation test on the ring at $N = 5$ cyclically shifts the qubit-to-player wiring and the whole per-player payoff vector permutes rigidly with it (deviation $\sim 10^{-16}$); and applying the same depolarizing weight to the *ideal final state* instead of per-gate collapses the per-player spread to ~ 0 . The disadvantage therefore follows the entangler’s *circuit position* (gate ordering), not the ideal state or the player identity.

A distinct graph-role asymmetry was tested on hardware. A wiring-permutation control on the star at $N = 4$ (Sec. VI-B4, Fig. 10C) reassigns players to different physical qubits and moves the measured per-player payoff vector by at most 0.035, inside the preregistered threshold, while the hub–leaf split itself (≈ 0.34 against ≈ 1.21) is unchanged. The ring simulation identifies sensitivity to circuit ordering. The star hardware control supports persistence of its hub–leaf asymmetry under the tested physical assignments; it does not identify the cause of the ring effect.

6) *The Finite-Round Adaptation Pilot:* An exploratory W-state pilot at $N = 4$ tested one provisional independent round-robin best-response rule. No tested setting produced a beneficial converged fairness repair: at $p = 0$ the dynamics reached the 50-round cap near mean 0.51 without converging; at $p = 0.02$ they entered a payoff limit cycle with lower mean and larger spread; and at $p = 0.05$ they converged with little change. Two robustness checks were then run on this pilot, both simulation-only. Replacing round-robin with *simultaneous* best response, which removes move order entirely, reproduces all three verdicts, so the $p = 0.02$ non-

convergence is a property of the noisy best-response map rather than of the update order. Extending to $N = 5$, $W N = 5$ reproduces the $N = 4$ verdicts, while ring $N = 5$ fails to converge at every noise level tested and does so in the opposite direction: it drives a perfectly fair $p = 0$ baseline apart while leaving mean welfare unchanged, i.e. it redistributes rather than shrinks. Because those ring endpoints are snapshots of non-convergent trajectories we report the direction only and not the magnitudes. Belief-based rules such as fictitious play remain untested, and the pilot remains exploratory pending game-theory sign-off.

7) *Alternative Phase and Compact Evaluation*: The incentive boundary of Sec. IV-G separates a phase-choice limitation from a player-count limitation. At $N = 4, 5, 6, 7$, the alternative profile's minimum ideal Hawk/Dove gaps are respectively 0.7500, 0.4180, 0.5000, and 0.3734, whereas the original profile has a profitable Hawk deviation at each size. These are ideal results; the earlier device experiments used the original phase. Hardware tests of the alternative strategy are reported in Sec. VI-B5. The unrestricted witness remains profitable for both branches.

The ideal GHZ circuit also permits compact classical evaluation. Write $c = \cos(\gamma/2)$, $s = \sin(\gamma/2)$, $A = \bigotimes_j U_j|0\rangle$, and $B = \bigotimes_j U_j|1\rangle$. Expanding the preparation and inverse in Eq. (7) gives

$$|\psi_{\text{out}}\rangle = c^2 A + iscB - iscX^{\otimes N} A + s^2 X^{\otimes N} B. \quad (28)$$

Thus the output is a sum of at most four product states, even for nonidentical local strategies, and its Schmidt rank across every bipartition is at most four. This is a classical representation bound for the ideal GHZ circuit, not a bound for every family or noisy implementation.

Per-player payoffs can be evaluated without enumerating outcomes using

$$\pi_j = \frac{V}{N} p(0^N) + V \mathbb{E} \left[\frac{x_j}{k} \mathbf{1}_{\{k>0\}} \right] - \frac{C}{N} p(1^N). \quad (29)$$

For each pair of product branches, $1/k = \int_0^1 z^{k-1} dz$ turns the middle term into a polynomial integral of degree $N-1$. Gauss-Legendre quadrature with $\lceil N/2 \rceil$ nodes integrates it exactly up to floating point. Prefix and suffix products evaluate all players at each node in linear work, giving quadratic contraction work once the quadrature nodes are available. Nodes are cached; the timing experiment warms that cache before measurement and does not time node generation. The implementation uses no 2^N probability array or dense entangler. Tests compare arbitrary-profile payoffs with the dense oracle through $N = 7$; the resource experiment extends evaluation to $N = 128$. Timings in Fig. 5 include memory instrumentation overhead and are single-machine measurements, not asymptotic speedup evidence. Checking a fixed three-strategy candidate needs $1 + 2N$ profile evaluations; enumerating all equilibria still requires a separate search.

For a fixed profile, we also replace heuristic unilateral optimization with a global local-unitary response calculation.

Quaternion quadratic-form methods already occur in two-player quantum-game analysis [19]; here we use the single-gate structure with fixed multiplayer opponents. Write $U(q) = q_0 I + i(q_1 X + q_2 Y + q_3 Z)$ with real $\|q\| = 1$. Linearity of circuit amplitudes in this single gate makes the payoff $q^T M_j q$ for a real symmetric 4×4 matrix. Four coordinate evaluations and six normalized pair sums determine M_j ; its largest eigenvalue is the best-response payoff and its eigenvector supplies a witness. Independent evaluations check the reconstruction and witness. This certifies a best response to fixed opponents in the specified circuit; noisy compilations that depend on the strategy would need the quadratic form validated separately.

8) *Sensitivity to Partial Conflict Costs*: The mean-welfare identity implies that analytic-baseline retention equals $1 - p(1^N)$. Uniform independent classical bits therefore retain $1 - 2^{-N}$, or 99.21875% at $N = 7$. Such randomization is not an equilibrium of the dominance game, so high retention alone cannot establish coherent cooperation or incentive compatibility. All output patterns with between one and $N - 1$ Hawks preserve the original total welfare.

We test the dependence on that payoff rule with a separate sensitivity game. Let $\ell \in [0, 1]$ and define total conflict loss

$$L_\ell(k) = C \left[(1 - \ell) \mathbf{1}_{\{k=N\}} + \ell \frac{k(k-1)}{N(N-1)} \right]. \quad (30)$$

All-Dove still pays V/N ; at $k > 0$ each Hawk receives $(V - L_\ell(k))/k$ and each Dove zero. The original game is $\ell = 0$, while $\ell = 1$ charges according to the fraction of Hawk pairs. Both endpoint payoffs and the $N = 2$ game are preserved. At $V > C$, Hawk remains strictly dominant since its payoff is positive against every nonempty Hawk set and is $V > V/N$ against all-Dove.

We retrospectively reweight saved raw fold-1 counts at $\ell = 0, 0.25, 0.5, 1$, without claiming new hardware execution or new equilibrium tests. For the $N = 3, \dots, 7$ extension run, retention at $\ell = 1$ is respectively 98.17%, 96.00%, 94.91%, 92.46%, and 93.23%, compared with 75% for uniform independent outcomes at every size. At $N = 7$ the corresponding raw player floor is 0.4993. Sampling intervals come from the same count distributions and exclude calibration drift.

B. Hardware Validation on IBM Heron r2

1) *Single-Point Validation at $N = 3$* : We ran the validated Q_3 circuit (entangler decomposed to 6 native CZ gates) with 4096 shots on `ibm_fez`; 3911/4096 shots landed in $|000\rangle$ (Fig. 6). The measured mean payoff is 1.3311, or 99.835% of the ideal payoff $4/3$; subtracting the analytic all-Hawk baseline $1/3$ gives measured advantage 0.9978, or 99.780% of the ideal advantage 1.0.

A product of the recorded readout fidelities (0.9960, 0.9961, 0.9962) and two-qubit fidelities (edge errors 0.17% and 0.23%, applied in the recorded four-plus-two CZ split) gives a heuristic ground-state probability of about 0.977, whereas the measured value is $3911/4096 = 0.955$. Under a fixed

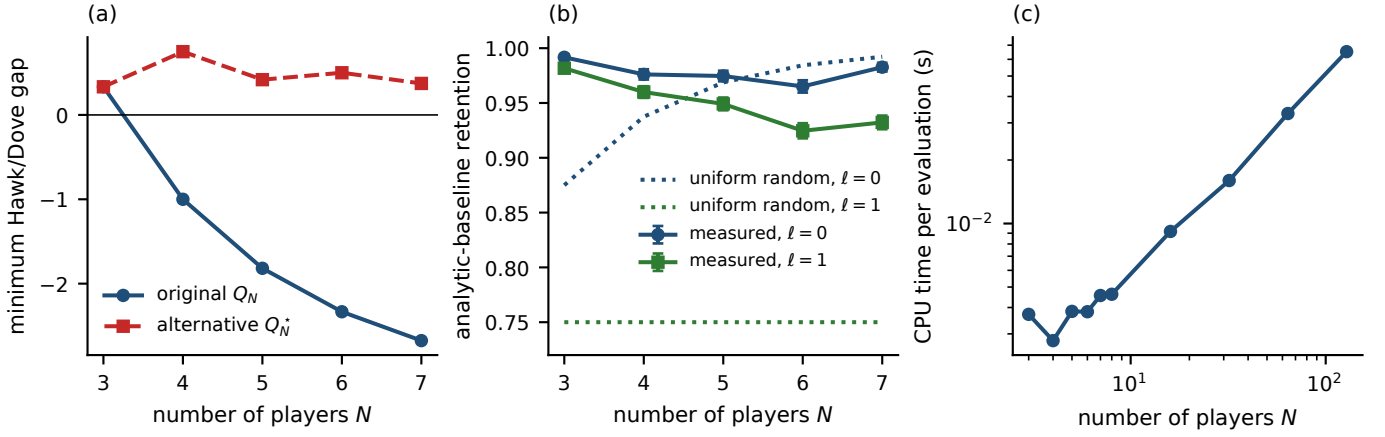


Fig. 5. Phase-branch, payoff-sensitivity, and compact-evaluation analyses. (a) Minimum ideal Hawk/Dove deviation gap for the original and alternative cooperative phase branches, each in its own restricted menu. (b) Retrospective raw fold-1 retention of the $N = 3-7$ hardware run under the original ($\ell = 0$) and partial-conflict ($\ell = 1$) payoff rules, with the values that uniformly random outcomes would give; error bars are 1.96 multinomial standard errors and exclude calibration drift. (c) CPU time of exact compact GHZ payoff evaluation through $N = 128$, a classical resource measurement.

binomial interpretation the difference is about 9.5 standard deviations, so the product has the right order of magnitude but substantially underpredicts non-ground weight; it is not a calibrated statistical model of this execution.

For every outcome with $0 < k < N$ Hawks, total payoff remains V , so the mean remains V/N . Only the all-Hawk outcome lowers total payoff to $V - C$:

$$\bar{\pi} = \frac{V}{N} - \frac{C}{N}P(1 \dots 1). \quad (31)$$

In the measured counts, 176 of the 185 shots outside $|000\rangle$ land on one- and two-Hawk outcomes, while only the 9 all-Hawk shots (0.22%) lower the mean. The identity reconstructs the measured payoff 1.3311 and advantage 0.9978. The mean-payoff observable is therefore first-order insensitive to single bit-flips from the ideal outcome: errors that flip one qubit redistribute payoff between players without lowering the mean payoff. This is the structural mechanism behind the scaling section’s central observation, that the advantage decays several times slower than the all-zero target-state population, and it is why the $N = 3$ advantage survives at 99.780% of its ideal while $P(|000\rangle)$ has already lost 4.5%.

2) Scaling at $N = 3, 4, 5$ with Error Mitigation:

All three N ran on prefixes of one pinned 5-qubit chain [59, 75, 74, 73, 79] (chosen by calibration error) in a single 11-circuit batch: two readout-calibration circuits plus $N \in \{3, 4, 5\} \times \text{CZ-fold } \lambda \in \{1, 3, 5\}$. Readout errors are corrected by tensored confusion-matrix inversion (constrained least squares) [8], [9]; zero-noise extrapolation folds each CZ locally ($\text{CZ} \rightarrow \text{CZ}^\lambda$, unitary-preserving since CZ is self-inverse) and extrapolates the readout-mitigated payoff linearly to $\lambda = 0$ [10], [11].

Calibration grouping and inclusion. The aggregate includes three executions on [59, 75, 74, 73, 79], with 4096 shots per circuit, spanning two provider calibration stamps: 2026-07-16

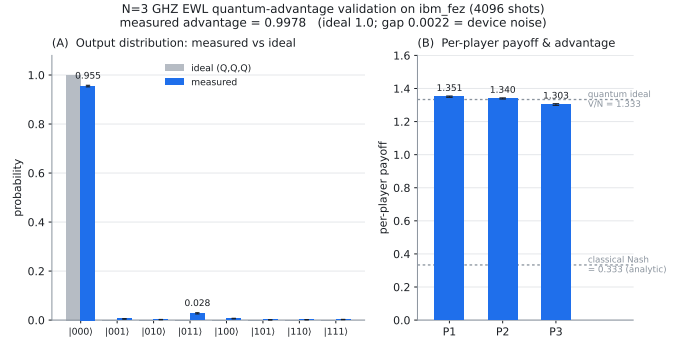


Fig. 6. Hardware validation of the $N = 3$ GHZ EWL quantum advantage on `ibm_fez` (IBM Heron r2). The validated (Q, Q, Q) circuit, with entangler J decomposed to 6 native CZ gates, was executed with 4096 shots. (A) Measured output distribution over the eight computational-basis states (blue) versus the noiseless ideal (grey); error bars are binomial shot noise. 3911/4096 shots (95.5%) fall in $|0 \dots 0\rangle$, the ideal (Q, Q, Q) output. (B) Measured per-player payoff (blue) against the analytic classical-Nash baseline (0.333, noiseless) and the quantum ideal $V/N = 1.333$; error bars propagate shot noise. The measured mean payoff 1.331 gives advantage 0.9978 vs the ideal 1.0. Physical qubits {136, 143, 142} (calibration 2026-07-15): readout error 0.38–0.40%, CZ error 136–143: 0.17%, 143–142: 0.23%.

08:30:13 +05:30 (two executions) and 2026-07-26 13:29:01 +05:30 (one execution). Executions sharing a stamp are averaged before computing the spread so that one calibration is not counted twice. A fourth execution on [20, 21, 22, 23, 24] is excluded because pooling it would confound calibration drift with a change of physical qubits. Table IV and Fig. 8 show the first execution alone.

Models and uncertainty. The ideal, device-model and depolarizing curves come from the first execution, not averages of refitted models. The depolarizing parameter $p_{\text{eff}} = 0.0018$ is fitted only at $N = 3$; its $N = 4, 5$ values are predictions. In Figs. 7 and 9, error bars are the sample standard deviation across the two calibration epochs, a coarse between-epoch spread from $n = 2$. They exclude within-run multinomial and

TABLE IV
MEASURED ADVANTAGE ON IBM_FEZ FOR THE FIRST EXECUTION (4096 SHOTS). REPEAT EXECUTIONS ARE DISCUSSED SEPARATELY; THE CROSS-CALIBRATION AGGREGATE OVER BOTH EPOCHS IS FIG. 7. IDEAL ADVANTAGE IS $3/N$; RETENTION IS ZNE/IDEAL. UNIFORMLY RANDOM OUTCOMES WOULD RETAIN $1 - 2^{-N}$: 87.5%, 93.8%, AND 96.9%.

N	ideal	raw	mitig.+ZNE	retention	$P(0\dots 0\rangle)$
3	1.00	0.9919	0.9938	99.4%	0.938
4	0.75	0.7386	0.7403	98.7%	0.922
5	0.60	0.5837	0.5887	98.1%	0.868

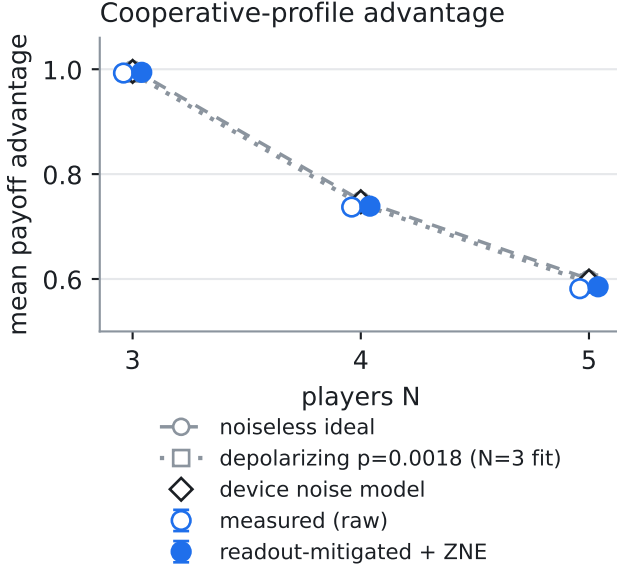


Fig. 7. Cooperative-profile payoff advantage on *ibm_fez* at $N = 3, 4, 5$, relative to the analytic noiseless classical-Nash baseline, comparing raw and mitigated-plus-ZNE measurements with ideal and noise-model curves. Error bars show the between-epoch sample standard deviation ($n = 2$ calibration epochs) and exclude within-run errors; depolarizing values at $N = 4, 5$ are predictions from the $N = 3$ fit.

weighted-fit standard errors. Figure 8 instead shows within-run payoff standard errors for the first execution. Among the plotted cases, $(Q, \dots, Q)$ passes the restricted pure-Nash check only at $N = 3$; $N = 4, 5$ measure a cooperative protocol, not an equilibrium.

The all-zero target population in Fig. 9 decays faster than the advantage: the mean-payoff observable is first-order insensitive to single bit-flips from $|0\dots 0\rangle$, since a one-Hawk outcome retains mean payoff V/N . The observable mechanism is developed in Sec. VII-A.

Two model predictions frame Table IV: the provider-calibrated device noise model reduced to the executed qubits, and a fit-one-predict-two test in our depolarizing model, with a single $p_{\text{eff}} = 0.0018$ fitted to the $N = 3$ mitigated-fold-1 point. For the first execution, the primary mitigated fold-1 endpoint misses at $N = 4$ by 0.0043 (2.14σ) and at $N = 5$ by

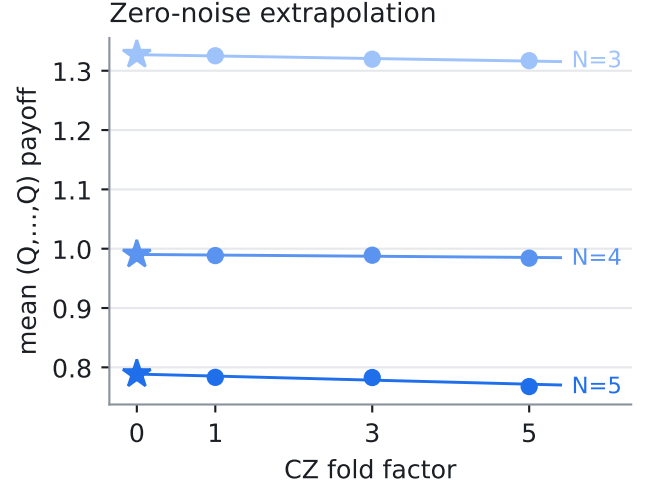


Fig. 8. Zero-noise extrapolation of the readout-mitigated mean payoff in the first execution at $N = 3, 4, 5$. Points at CZ fold factors 1, 3, 5 carry within-run payoff standard errors; weighted linear fits extrapolate to zero fold factor (stars).

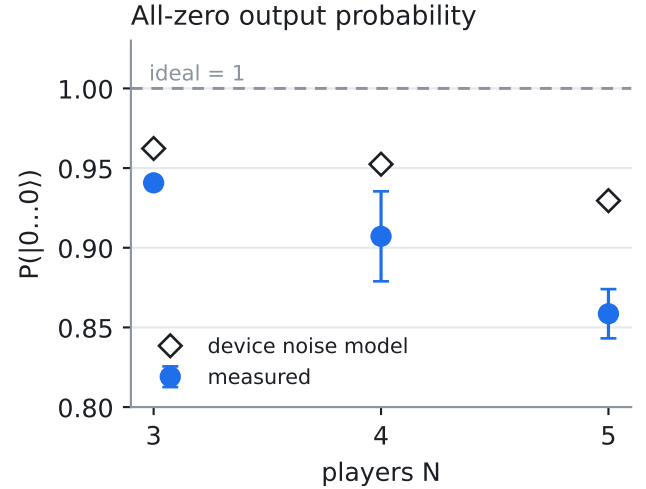


Fig. 9. All-zero output probability on *ibm_fez* at $N = 3, 4, 5$, compared with the device noise-model prediction and ideal target. Measured points average calibration epochs; error bars are the cross-epoch sample standard deviation ($n = 2$) and exclude within-run errors.

0.0101 (5.17σ). These residuals are retrospective, because the predictions were fixed after that execution. For the displayed ZNE endpoint, the corresponding misses are 0.0041 (2.07σ) and 0.0060 (3.05σ). Three preregistered repeats followed. In the first, under the same provider calibration, the conditional $N = 4$ endpoint passes ($z = -0.93$) and the $N = 5$ endpoint misses by 0.0092 (4.74σ). The other two, under new calibrations, miss at both sizes: on the different chain $[20, \dots, 24]$ by 0.0046 (2.31σ) at $N = 4$ and 0.0094 (4.83σ) at $N = 5$, and at the 2026-07-26 epoch by 0.0113 (5.68σ) and 0.0186 (9.55σ). The $N = 4$ endpoint therefore passes

in one of three repeats. The repeated $N = 5$ deficit sign is observational, not a preregistered sign test.

Of five preregistered baseline models, the CZ-exponential law has the smallest summed z^2 in all four executions, including both cross-calibration repeats. It nevertheless misses $N = 5$ outside the shared 95% comparison yardstick in every run; that yardstick is for ranking, not a model-specific calibrated interval. That ranking is specific to $N \leq 5$: in the $N = 6, 7$ extension (Sec. VI-B3) the CZ-exponential law loses to the depolarizing model at $N = 7$ and both fail their tests, so the ranking is not a physical scaling law. The four executions span three distinct provider calibrations, the minimum of the planned three to five, but one of those calibrations is represented only by the [20, ..., 24] execution that the aggregate excludes; the chain-matched sample covers two calibrations. The $N = 5$ deficit has the same sign in all three repeats. Its magnitude is not stable, however: it is 0.0092 under the source run's calibration, 0.0094 on the other chain, and 0.0186 at the 2026-07-26 epoch, so the variation across calibrations is itself of order the effect. Model-specific predictive uncertainties remain needed.

3) *Extension to $N = 6, 7$* : A later execution extends the curve to $N = 3, \dots, 7$ on one pinned seven-qubit chain [97, 107, 108, 109, 110, 111, 98] (17 circuits, 20 QPU-seconds). The chain was fixed with the predictions before submission, so the executed qubits match the ones the predictions describe. Measured mitigated fold-1 advantage retains 99.2%, 97.5%, 97.3%, 96.3% and 98.1% of the ideal $3/N$ at $N = 3, \dots, 7$ while $P(|0\dots 0\rangle)$ falls from 0.943 to 0.784. Uniformly random outcomes would retain $1 - 2^{-N}$, that is 87.5%, 93.8%, 96.9%, 98.4% and 99.2%. The measured values exceed that comparator through $N = 5$ but fall below it at $N = 6, 7$, where the device places more weight on the all-Hawk outcome than independent fair coins would. High retention at $N = 6, 7$ is therefore not evidence of coherent cooperation.

Two competing one-parameter models were preregistered, both fitted on the same $N = 3$ point and extrapolated with no $N > 3$ information: the depolarizing p_{eff} model and the CZ-exponential retention law $A_N = A_N^{\text{ideal}} r^{\text{CZ}N}$. Both failed the preregistered 1.96σ test at $N = 6, 7$, and they failed differently: the depolarizing model over-predicts at every N , while the CZ-exponential model over-predicts at $N = 4, 5, 6$ and then under-predicts at $N = 7$. A one-parameter retention law fitted at $N = 3$ cannot reproduce that curvature.

The comparison also split: the CZ-exponential law is closer at $N = 4, 5, 6$ but the depolarizing model is closer at $N = 7$. Measured advantage is strictly decreasing in N , as predicted, but *retention* is not: $N = 7$ keeps more of its ideal advantage than $N = 4, 5, 6$ despite carrying the most gates and the lowest target-state population. Part of this is built into the metric: retention equals $1 - p(1^N)$, and even uniformly random outputs land on the single all-Hawk outcome less often as N grows. The $N = 7$ ZNE estimate, however, overshoots the

TABLE V
MULTI-TOPOLOGY BATCH ON IBM_FEZ (4096 SHOTS, ONE PINNED QUBIT SET). RETENTION IS MEASURED MITIGATED FOLD-1 ADVANTAGE AS A FRACTION OF THAT CELL'S OWN NOISELESS ADVANTAGE, WHICH IS NOT $3/N$ FOR EVERY TOPOLOGY.

topology	N	CZ	ideal adv	measured	retention
GHZ	3	6	1.0000	0.9886	98.9%
GHZ	4	9	0.7500	0.7425	99.0%
GHZ	5	14	0.6000	0.5921	98.7%
W	3	42	1.0000	0.9792	97.9%
ring	3	10	1.0000	0.9899	99.0%
ring	4	20	0.0000	0.0894	n/a
ring	5	28	0.6000	0.5960	99.3%
star	3	5	1.0000	0.9934	99.3%
star	4	9	0.7500	0.7423	99.0%
star	5	17	0.6000	0.5977	99.6%
fully-conn.	3	10	1.0000	0.9887	98.9%
fully conn.	4	31	0.7500	0.7389	98.5%

noiseless ideal (0.4437 against 0.4286), which no physical noise model can produce. We report that overshoot as an unexplained observation rather than a result. Distinguishing payoff geometry at odd N from an end-of-chain qubit effect or an extrapolation artefact requires a repeat of *this* $N = 3 \dots 7$ batch specifically: the cross-calibration repeats aggregated in Fig. 7 re-execute the $N = 3, 4, 5$ batch and so cannot address it, and no such $N = 6, 7$ repeat has been run.

4) *Topology, Equilibrium and Fairness on Device*: The experiments above are all GHZ. To put the paper's topology, equilibrium and fairness claims on device data we ran a single 49-circuit batch (4096 shots, 55 QPU-seconds) spanning five entangler topologies, unilateral strategy deviations, wiring permutations and an entanglement-angle sweep, with all predictions preregistered. Cells were selected from *measured* routed CZ counts on the real coupling map rather than from pre-routing gate counts: on heavy-hex hardware the two differ sharply, and fully connected $N = 5$ (67 CZ) and W $N = 4, 5$ (107, 219) were excluded on that basis.

The eleven cells with nonzero ideal advantage retain 97.9–99.6% of that value (Table V, Fig. 10A), above the 87.5–96.9% that uniformly random outcomes would retain at $N = 3$ –5. They include W $N = 3$ at 42 CZ, seven times the gate count of GHZ $N = 3$. High retention is therefore not specific to GHZ: ring $N = 5$ and star $N = 5$ sit at $P(|0\dots 0\rangle) \approx 0.10$ –0.11 while retaining over 99% of their analytic-baseline advantage. Their ideal outputs are not generally all-zero, so these populations alone do not measure preparation error or fidelity to their own ideal states, and these cells do not test the single-flip mechanism of Sec. VII-A. The topology axis does not separate these systems by how much advantage they retain; it separates them by *structure*, as the next two results show.

a) *Hawk-deviation tests on hardware*: At GHZ $N = 3$ the three unilateral Hawk deviation gaps measure +0.2895, +0.2726 and +0.3512 (Fig. 10B), all positive and many standard errors clear of zero. This preregistered test passed

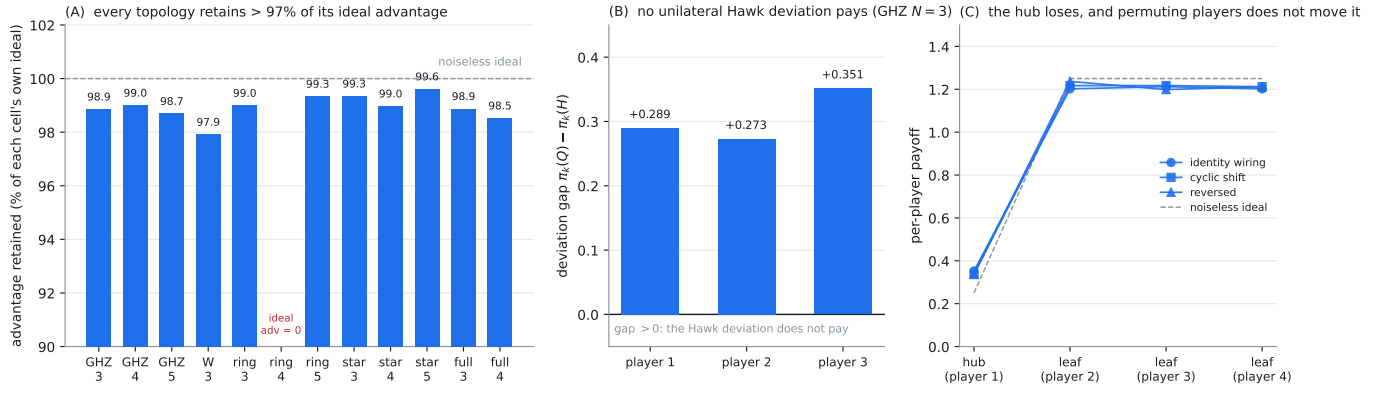


Fig. 10. Multi-topology EWL batch on `ibm_fez`: 49 circuits $\times$ 4096 shots on one pinned qubit set, with all predictions preregistered. (A) Measured mitigated fold-1 advantage as a percentage of each cell’s *own* noiseless advantage; the eleven nonzero-baseline cells retain 97.9–99.6%. ring $N = 4$ carries no percentage because its noiseless advantage is exactly zero: its ideal circuit is deterministic on $|1 \dots 1\rangle$, so the $+0.089$ it measures is created by decoherence, and zero-noise extrapolation reduces it to $+0.037$. (B) Unilateral Hawk deviation gaps $\pi_k(Q, Q, Q) - \pi_k(H \text{ at } k)$ at GHZ $N = 3$: all three are positive, so no measured Hawk deviation pays; Dove deviations were not measured. (C) star $N = 4$ per-player payoffs under three wirings. The hub earns ≈ 0.34 while each leaf earns ≈ 1.21 , and permuting players across physical qubits does not move the pattern: the disadvantage is locked to the graph position, not to the qubit.

at every deviating position: the (Q, Q, Q) profile resists the three measured Hawk deviations. Dove deviations were not part of this batch; the all-player tests of Sec. VI-B5 include them.

A preregistered sweep of the same gap in the entanglement angle changes sign as predicted: -0.6877 at $\gamma = 0.30\pi$ and $+0.2467$ at $\gamma = 0.45\pi$. The intermediate point $\gamma = 0.40\pi$ was declared marginal in advance and excluded from the pass criterion; it returned $+0.0067$, statistically indistinguishable from zero. The measured Hawk-deviation incentive therefore changes sign with entanglement strength, with the transition bracketed near $\gamma = 0.4\pi$ at $N = 3$.

b) Position-locked unfairness is a property of the topology: star $N = 4$ is maximally unequal even noiselessly: the analytic per-player vector is $(0.25, 1.25, 1.25, 1.25)$, the hub being the loser. On device the measured vector is $(0.353, 1.202, 1.212, 1.202)$, and under two wiring permutations that reassign players to different physical qubits it moves by at most 0.035 (Fig. 10C), inside the preregistered threshold. The null hypothesis of wiring invariance is therefore not rejected: the disadvantaged position is determined by the entanglement topology, not by which physical qubit a player was allocated. This supports the star graph-role effect under the tested assignments, separately from noise-induced ring circuit-order asymmetry.

c) A cell whose advantage is made of noise: ring $N = 4$ is the one cell whose noiseless circuit is deterministic on $|1 \dots 1\rangle$, the all-Hawk outcome, so its ideal advantage is exactly zero. It measured $+0.089$, the smallest of the twelve cells but clearly positive, and zero-noise extrapolation reduced it to $+0.037$; no uncertainty was recorded for that extrapolated value. All three parts of that outcome were predicted in advance. A pipeline that merely inflated whatever it touched could not produce a cell where decoherence *creates* apparent

advantage and the extrapolation reduces it.

d) What did not work: The preregistered per-cell predictions of *absolute* advantage failed, in both the raw device-noise-model form and a CZ-linear bias-corrected form, and that comparison is additionally confounded: the provider recalibrated between the predictions and submission and the qubit selector moved the pinned set, so those z -scores mix model error with a change of qubits. The structural results above do not depend on those predictions. The deviation, sweep, and wiring results are within-job differentials: a payoff difference between two profiles measured on the same qubits in the same job, or a permutation against its own twin. The ring cell is ranked against the other eleven in the same job, and retention compares each measurement with its own noiseless ideal rather than with a model prediction. Absolute-magnitude prediction of NISQ payoffs from a device noise model remains out of reach here; the structural claims do not depend on it.

5) Alternative-Phase Deviation Tests: This experiment evaluates Q_N^* separately from the original phase. At $N = 3$, $Q_3^* = Q_3$, so that size retests the original phase with Dove deviations added. On 7 September 2026, a preregistered plan fixed a seven-qubit chain, $[141, 142, 143, 144, 145, 146, 147]$, on `ibm_fez` and used its prefixes for $N = 3, \dots, 7$. An 18-circuit pilot run preceded the complete 70-circuit batch. The complete batch contains a cooperative profile at each size, both Hawk and Dove deviations for every player, original-phase controls, an unrestricted counterstrategy, and two readout calibrations. Every circuit uses 4096 shots. Each circuit is checked against its dense operator and its compiled ideal output, and the batch is rehearsed on a device noise model before submission.

The primary analysis uses raw implemented payoffs: calibration circuits are diagnostic and no readout correction or ZNE is applied to these tests. With $M = 50$ primary

deviations and $\alpha = 0.05$, the simultaneous lower bound on each cooperative-minus-deviation gap is

$$g_j^L(a) = \hat{g}_j(a) - \sum_{s \in \{\text{coop}, \text{dev}\}} V \sqrt{\frac{\log(2M/\alpha)}{2n_s}}, \quad (32)$$

where $n_s = 4096$. Hoeffding's inequality and a union bound cover the two one-sided mean errors for every gap. These bounds assume independent shots from stationary circuit distributions; they do not cover temporal drift. The preregistered approximate-equilibrium tolerance is $-g_j^L(a)/(V/N) \leq 0.05$ for every deviation of every player. The overall test requires all five sizes to pass.

In the first complete batch, every measured gap is positive, but the simultaneous criterion passes only at $N = 3, 4, 5, 6$. The minimum observed gaps are 0.2360, 0.5677, 0.2157, 0.2627, and 0.0109, respectively. The corresponding maximum upper bounds on normalized deviation gain are 0.0058, -0.3240 , 0.0349, -0.0285 , and 0.4074. Thus the overall test fails: a positive point gap at $N = 7$ does not establish the requested tolerance. At $N = 4, 6$ the simultaneous lower gaps are strictly positive; at $N = 3, 5$ the supported statement is approximate equilibrium at the declared tolerance.

This all-player test locates an implementation boundary absent from mean welfare alone. At $N = 7$, target population is 0.7495 and analytic-baseline retention is 0.9600, below the 0.9922 that uniformly random outcomes would retain, and incentive compatibility is unresolved. The unrestricted counterstrategy, run as a diagnostic, gains between 2.4604 and 2.7052 over cooperation for player zero across the five sizes, consistent in direction with the analytic counterexample.

A second complete batch reused the same circuits, shots and decision rule. Each batch is judged separately; their counts are not pooled to rescue a failed criterion. The repeat supports the criterion at $N = 4, 6$, but not at $N = 3, 5, 7$ (Table VI). All 50 point gaps are positive in the first batch; one is slightly negative in the repeat, at $N = 7$. This is not a statistically established profitable deviation. Both batches fail the overall five-size test, while the positive lower gaps at $N = 4, 6$ reproduce. The executions occurred minutes apart on the same chain; this is a within-session repetition, not evidence across independent calibration epochs. The pilot and two complete batches consumed $21 + 77 + 77 = 175$ seconds of provider-reported charged QPU time.

VII. DISCUSSION

A. Why the Advantage Outlives the State

Target-state population decays several times faster than the advantage (Table IV): a single bit-flip from $|0 \dots 0\rangle$ yields a one-Hawk outcome whose *mean* payoff is still exactly V/N , so the mean-payoff observable is first-order insensitive to uncorrelated single flips. Readout mitigation consequently moves the mean very little; its per-player effects are mixed and remain diagnostic analyses rather than preregistered acceptance tests.

TABLE VI
ALTERNATIVE-PHASE ALL-PLAYER TESTS, JUDGED SEPARATELY IN EACH BATCH. $g_{\min}$ IS THE SMALLEST OBSERVED PAYOFF GAP; $u_{\max}$ IS THE LARGEST SIMULTANEOUS UPPER BOUND ON DEVIATION GAIN NORMALIZED BY V/N . THE PREREGISTERED TOLERANCE REQUIRES $u_{\max} \leq 0.05$.

N	First batch		Repeat	
	$g_{\min}$	$u_{\max}$	$g_{\min}$	$u_{\max}$
3	0.2360	0.0058	0.1362	0.0806
4	0.5677	-0.3240	0.5716	-0.3279
5	0.2157	0.0349	0.1861	0.0719
6	0.2627	-0.0285	0.2483	-0.0069
7	0.0109	0.4074	-0.0033	0.4322

The displayed ZNE estimates are higher for every series, but no recorded population analysis attributes that shift specifically to suppression of CZ-created all-Hawk weight. The same identity limits what retention can show: uniformly random outcomes retain $1 - 2^{-N}$ (Sec. VI-A8), and the $N = 6, 7$ extension falls below that comparator (Sec. VI-B3). First-order insensitivity explains why retention degrades slowly; it does not make high retention evidence of coherent cooperation. The structural robustness is a property of the game's payoff geometry, not of the device.

The same single-flip terms that leave the mean invariant are what create winners and losers. Under the all-Dove branch each one-Hawk outcome transfers V to a single player and zero to the rest: the mean is protected, the individual player is not, and under gate-level noise the disadvantaged position is locked to the entangler's circuit layout rather than to any coordination failure (Sec. VI-A4). In the single exploratory W-state $N = 4$ finite-round pilot, the provisional independent round-robin best-response rule produced no beneficial converged fairness repair in any tested setting. At $p = 0$ the dynamics reached the 50-round cap near mean 0.51 without converging; at $p = 0.02$ they entered a payoff limit cycle with lower mean and larger spread; and at $p = 0.05$ they converged with little change. The verdict survives both checks we applied to it: it is unchanged under simultaneous rather than round-robin updates, and W $N = 5$ reproduces it. Ring $N = 5$ does not converge at any noise level tested and fails in the opposite direction, spreading a perfectly fair baseline apart at unchanged mean welfare, reported as a direction rather than a magnitude, since those endpoints are non-convergent snapshots. No belief-based rule has been tested. The advantage we report can therefore coexist with unequal per-player outcomes; the per-player floor is a relevant diagnostic for mechanism design.

B. Scalability and Practicality

Scalability has distinct mathematical, computational, and physical meanings in this protocol. The alternative phase branch has an ideal restricted-menu equilibrium for every N , and compact GHZ evaluation reaches $N = 128$ without dense state enumeration. Neither result increases the number

of simultaneous players demonstrated on hardware: the native device evidence ends at $N = 7$. The alternative-phase batches test every D/H deviation but do not resolve the tolerance at $N = 7$. Ideal per-player gains scale as C/N , incentive margins decrease with N , and the permitted entanglement-angle interval narrows. Practical scaling therefore requires explicit statistical precision and implementation budgets.

The protocol assumes a shared preparation stage and joint inverse/readout. Our centralized experiments do not demonstrate a distributed quantum market or remove the need to trust preparation, decoding, and enforcement of a permitted strategy menu. Market architectures motivate interpretations of the graph families, but are not validated deployment models. Comparisons with classical mechanisms must state which mediation and enforcement resources are available [20].

C. Limitations and Future Work

Our primary game fixes $V = 4$ and $C = 3$ in a strict-dominance regime and charges conflict cost only at the all-Hawk outcome. High mean retention is therefore compatible with random outcomes, unequal allocations, and profitable deviations. The partial-conflict sensitivity analysis tests this modeling choice retrospectively; it does not establish a realistic economic payoff model or a noisy equilibrium of the altered game.

All restricted equilibria refer to explicitly named menus. Both cooperative phase families admit a profitable unrestricted SU(2) deviation. Original-phase device measurements tested Hawk deviations only, and do not establish resistance to Dove deviations under hardware noise. Arbitrary quantum channels, ancillary systems, coalitions, and adaptive multi-round strategies are outside our certification scope. Symmetric strategy search also does not exclude better asymmetric profiles or other equilibria.

Entangler-family comparisons depend on the chosen generator, interpolation angle, compiled circuit, routing, and noise placement. The W results concern one specified operator; equal nominal angles do not equalize interaction budget or entangling power. The normalized interaction convention defined above remains an unexecuted sensitivity control. Noiseless star graph-role asymmetry and noise-induced ring circuit-order asymmetry are distinct effects; the star wiring experiment alone does not identify the ring's mechanism.

Depolarizing noise is a controlled model rather than a complete device description. Readout correction and zero-noise extrapolation introduce assumptions and uncertainty; extrapolated estimates can exceed physical payoff bounds. Chain-matched scaling uncertainty rests on two calibration epochs, and the $N = 6, 7$ batch has not been repeated. Preregistered one-parameter models did not extrapolate reliably. The complete alternative-phase batches are repetitions within one session, and fail their overall five-size criterion; only $N = 4, 6$ pass in both. The compact simulation results do not extend the demonstrated physical player count. The

finite-round adaptation pilot remains exploratory and does not establish convergence of general learning rules.

VIII. NOVELTY AND TECHNICAL DIFFERENTIATION

Prior multiplayer EWL studies and trading experiments establish the setting in which this work belongs [2], [3], [6], [7]. Our contribution is the joint accounting of restricted incentives, payoff geometry, and implementation-dependent fairness in a specified allocation game. The phase-branch boundary explains why the original cooperative strategy fails beyond three players and provides a different ideal restricted-menu profile, accompanied by an unrestricted counterexample.

Matched compilation controls restrict how much of a measured family ordering can be attributed to ideal correlation structure. The payoff sensitivity analysis likewise restricts how much high mean retention establishes about cooperation. The hardware results include both positive differential tests and unsuccessful quantitative predictions. These components support a study of incentive and observable robustness; they do not establish computational quantum speedup or an operational quantum market.

DATA AND CODE AVAILABILITY

The simulation code, hardware-execution code, serialized circuits, raw hardware counts, calibration records and all analysis outputs are available at <https://github.com/QuantumCompTeam/Multiplayer-QHD-Game>.

IX. CONCLUSION

We studied how entangler family, implementation noise, and player count affect welfare and incentives in a multiplayer EWL allocation game. The original GHZ cooperative phase Q_N has a restricted-menu equilibrium only at $N = 2, 3$, while the alternative branch Q_N^* , which coincides with Q_N at $N = 2, 3$, satisfies the ideal restricted condition for every $N \geq 2$ at maximal entanglement. Both profiles admit an explicit profitable unrestricted SU(2) deviation. An all- N restricted existence statement therefore coexists with an exact strategy-access limitation.

Original-phase hardware experiments evaluate the cooperative protocol through seven players; at $N = 3$ the multi-topology batch finds no profitable measured Hawk deviation. Two complete alternative-phase batches, which also test Dove deviations, support the preregistered incentive criterion at $N = 4, 6$ in both runs. The $N = 3$ criterion, where $Q_3^* = Q_3$, and the $N = 5$ criterion pass only in the first batch, and $N = 7$ remains unresolved despite high mean retention, which at $N = 6, 7$ falls below the uniformly random comparator. Both batches fail the overall five-size test. High mean retention is partly built into the all-Hawk-only conflict penalty; the uniform comparator and retrospective partial-conflict sensitivity make that dependence explicit. At $N = 7$, the partial-conflict model retains 93.23% in raw data against

a 75% uniform comparator, without implying an equilibrium or a computational advantage.

Compact classical evaluation through $N = 128$ provides a scalable analysis tool, while compilation controls and calibration records delimit the physical evidence. Repetition across independent calibration epochs and explicit resource assumptions remain necessary to connect ideal incentive compatibility with a reliable implementation.

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